

The critical correlation exponent of the planar XY model

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Abstract

For the ordinary nearest-neighbor XY model on the square lattice, we prove the Berezinskii–Kosterlitz–Thouless prediction that the critical spin-correlation exponent is $1/4$. More precisely, at the mass-defined critical inverse temperature, the infinite-volume correlation is $n^{-1/4+o(1)}$. The infinite-volume limit through free square boxes is taken before the separation limit.

1 Introduction

The two-dimensional XY model has an algebraic phase even though its spins have no preferred direction. The Berezinskii–Kosterlitz–Thouless theory predicts that the correlation exponent at the boundary of that phase is exactly $1/4$ [2, 7, 6]. This paper proves that prediction for the nearest-neighbor cosine interaction on the square lattice, at the threshold defined by vanishing horizontal mass. The conclusion concerns free-box infinite-volume correlations at all integer separations.

Several earlier results establish the surrounding phase picture. McBryan and Spencer proved power-law upper bounds for planar spin correlations [11]. Fröhlich and Spencer proved the existence of the low-temperature algebraic phase by a Coulomb-gas analysis [4]. Van Engelenburg and Lis obtained the exponential-versus-polynomial dichotomy, including a polynomial lower bound at the threshold, by directed currents and dual heights [14]. Their critical lower bound is the external endpoint input used here; its exponent is not sharp.

The dual-height approach also has a substantial independent history. Aizenman, Harel, Peled, and Shapiro developed integer-restricted Gaussian and annealed Gaussian methods for depinning and BKT phases, including a Gaussian-mixture representation of Bessel weights [1]. Lammers identified the exact relation between XY mass and its dual height mass [8], and proved a qualitative localization dichotomy with a positive universal height-variance gap [9]. The broader duality identities of van Engelenburg and Lis [15] and the surface-tension derivative formula of Durand and Lammers [3] further connect spin decay and height fluctuations. These results do not identify the endpoint coefficient or its magnetic correlation exponent. Those two identifications are the tasks addressed below.

Our companion article [12] supplies the finite Gaussian comparisons, annular geometry and Gaussian estimates, and local analytic maps used in the proof. Its discrete-height theorems concern quadratic weights, and its spin theorem concerns sufficiently low fixed temperatures. We therefore prove the model-specific transfers to the Bessel interaction in this paper. Every companion citation gives a numbered statement or equation. Section 1.3 specifies the hypotheses at these interfaces.

Three extensions are particularly important. First, a subdivision by lattice Gaussian chains transfers finite comparisons to Bessel weights, but not their bare Gaussian variance bound. Unit-edge flow costs and arbitrarily high polynomial moments replace that bound, including after cable

exploration and for very small pin probabilities. Second, convergence of ordinary height observations does not determine a winding-sector partition ratio. A finite isolation expansion on geometric annuli supplies that comparison, with only a small error per annular scale. Third, an additional positive form on *averages*, rather than on microscopic heights, supplies the large-field reserve needed for an open Gaussian regime. This form is removed by a variance comparison before drawing a conclusion about the original model.

1.1 Model and statement

For an integer $L \geq 1$, put

$$\Lambda_L = [-L, L]^2 \cap \mathbb{Z}^2, \quad E_L = \{\{x, y\} \subset \Lambda_L : |x - y| = 1\}.$$

Every edge occurs once. At inverse temperature $b > 0$, the angles $\theta_x \in \mathbb{R}/(2\pi\mathbb{Z})$ have law

$$d\mu_{L,b}(\theta) = Z_{L,b}^{-1} \exp \left\{ b \sum_{\{x,y\} \in E_L} \cos(\theta_x - \theta_y) \right\} \prod_{x \in \Lambda_L} \frac{d\theta_x}{2\pi}. \quad (1.1)$$

There are no exterior edges, fixed spins, fields, or boundary identifications. With $e_1 = (1, 0)$, set

$$C_b(n) = \lim_{\substack{L \rightarrow \infty \\ L \geq n}} \mathbb{E}_{\mu_{L,b}} \cos(\theta_0 - \theta_{ne_1}), \quad m(b) = \lim_{n \rightarrow \infty} -\frac{1}{n} \log C_b(n), \quad (1.2)$$

and

$$b_c = \inf\{b > 0 : m(b) = 0\}. \quad (1.3)$$

The correlation limit exists by ferromagnetic coupling monotonicity [5]. The multiplicative correlation inequality [14, arXiv version 2, Lemma 19], after free exhaustion and translation invariance, gives $C_b(n+m) \geq C_b(n)C_b(m)$. Fekete's lemma therefore gives the mass limit. The dichotomy in [14, Theorem 1] implies $0 < b_c < \infty$ and includes b_c in the massless regime. The parameter b is exactly the β of the usual cosine convention. Throughout the proof, the first limit in (1.2) is taken before the separation limit.

Theorem 1.1. *For the model (1.1) and the threshold (1.3),*

$$\lim_{\substack{n \rightarrow \infty \\ n \in \mathbb{N}, n \geq 2}} \frac{\log C_{b_c}(n)}{\log n} = -\frac{1}{4}.$$

Equivalently, for every $\varepsilon > 0$ there is $n_0 < \infty$ such that

$$n^{-1/4-\varepsilon} \leq C_{b_c}(n) \leq n^{-1/4+\varepsilon} \quad (n \geq n_0).$$

We use the following established, nonsharp critical input.

Proposition 1.2 (Critical polynomial lower bound). *For every $n \geq 1$,*

$$C_{b_c}(n) \geq \frac{1}{8n}.$$

Proof. Theorem 1 of van Engelenburg and Lis [14] is stated for free-boundary XY spins with density $\exp\{\frac{b}{2} \sum_{\{x,y\}} (\sigma_x \overline{\sigma_y} + \overline{\sigma_x} \sigma_y)\}$. Writing $\sigma_x = e^{i\theta_x}$ gives exactly (1.1). Their infinite-volume correlation is the free finite-graph exhaustion, hence agrees with (1.2); reflection of all angles removes its imaginary part. Their theorem gives exponential decay below a threshold and the lower bound $1/(8|x-y|)$ at and above it. These alternatives have positive and zero horizontal mass, respectively, so their threshold is (1.3). Specialize to $x = 0, y = ne_1$. $\square$

1.2 Structure of the proof

Section 2 constructs the free height coefficient $a = a(b)$, proves $a \in \{0\} \cup [8\pi, \infty)$, and gives the spin upper exponent $-2\pi/a$, with arbitrarily fast polynomial decay when $a = 0$. Section 3 proves the mixed-boundary and conditional pin limits needed for localization. Section 4 proves the matching lower exponent whenever $a > 0$, and thus zero spin mass in that case. Section 5 shows that $a(b) > 8\pi$ implies positivity of the coefficient in a neighborhood of b . Finally, Section 6 combines these facts with Proposition 1.2 to identify $a(b_c) = 8\pi$. No continuity of the unknown function $a(b)$ is assumed.

The constants can be followed through a short chain of identities. The height period is 2π , and a radial unit flow has energy $(2\pi)^{-1} \log r + O(1)$. Summing its Fourier sectors yields the gap $a \geq 8\pi$ whenever $a > 0$. A pair of opposite angular singularities has energy $4\pi \log n + O(1)$, giving spin exponent $2\pi/a$. The critical lower bound excludes $a(b_c) = 0$, while openness excludes $a(b_c) > 8\pi$. Thus $2\pi/a(b_c) = 1/4$.

Convention or limit	Meaning in this paper
Spin coupling	Each unoriented nearest-neighbor edge appears once in $b \cos(\theta_x - \theta_y)$.
Dual height	$h \in 2\pi\mathbb{Z}$, with normalized edge weights $p_b(j) = e^{-b} I_j(b)$. The ordinary unit-edge Laplacian is A .
Effective coefficient	Height covariance is aA^{-1} ; the corresponding Gaussian energy is $\langle h, Ah \rangle / (2a)$.
Thermodynamic limit	First $L \rightarrow \infty$ at each fixed integer n ; then $n \rightarrow \infty$, including nondyadic separations.
Subdivision	The chain length tends to infinity on each fixed primary graph before a macroscopic geometry is refined.
Bands and magnetic holes	Bands use mesh refinement, then band-cell refinement, then increasing band strength. Magnetic hole size and localization parameters are fixed through the separation limit.

Table 1: Normalizations and the order of limits. A Gaussian comparison at an auxiliary scale does not change the bare XY coupling.

1.3 Companion inputs and Bessel transfers

The quadratic companion is used at three distinct levels. First, [12, Lemma 2.1] applies on finite-dimensional lattices with positive quadratic precision. In Lemma 2.2 we put the Bessel law on finite Gaussian-chain approximants and pass to the limit with every primary graph fixed. Affine supports, centered constraints, observation penalties, and numerator-only real tilts are checked there. Lemma 2.3 supplies uniform moment bounds in the ordinary unit-edge flow norm. The bare inverse precision of a subdivided chain is not used as a uniform bound.

Second, [12, Lemma 3.12] and [12, Proposition 3.14] concern fixed face-connected rectangular geometries, ordinary real Gaussian forms, observation-cell masses, and separated pins. These geometric and Gaussian conclusions apply with the coefficient $a(b) > 0$. The Bessel mixed-boundary and conditional statements are instead proved in Theorems 3.1 and 3.2: primary path bounds, polynomial cluster moments, and a relative small-tilt estimate replace the quadratic tail argument. Full conditional limits refer to typical guard data through finite probes; they are not uniform limits at every microscopic boundary value. Lemma 3.9 checks that all restored supports lie in the pinned core.

Third, the analytic map [12, Proposition 8.3] acts on small real, even, periodic activities with block and square covariance, in the full analytic norm with enlarged support collars and regulator reserve. Lemma 5.4 verifies every such condition for the strengthened Bessel law, at a fixed gapped

Gaussian reference $a' > 8\pi$. Its entry proof is given here; the companion's quadratic physical entry theorem is not an assumption about Bessel weights. The subsequent Gaussian-curve argument uses a common small disk for sufficiently large starting scales, with the reference frequency held fixed. Finally we remove the added precision by comparison. This gives the openness needed at criticality without any continuity assumption on $a(b)$.

2 Dual heights and the structural coefficient

Throughout the proof, $b > 0$ denotes the spin inverse temperature. The height spacing is 2π . On an original, or *primary*, dual edge the weight of the difference $2\pi j$ is

$$p_b(j) = e^{-b} I_j(b), \quad \sum_{j \in \mathbb{Z}} p_b(j) e^{tj} = \exp\{b(\cosh t - 1)\}.$$

Thus p_b is the law of the difference of two independent Poisson variables of mean $b/2$. An unoriented edge is counted once. For a retained primary graph, A is its unit-conductance Laplacian:

$$\langle g, Ag \rangle = \sum_{\{x,y\}} (g_x - g_y)^2.$$

This normalization converges to $\int |\nabla g|^2$. Pairings on a finite graph use counting measure. A free connected height component is taken modulo common translations by $2\pi\mathbb{Z}$; one may choose a pinned representative when evaluating translation-invariant tests. When confining observations control the constant, all its translation copies are instead summed.

2.1 Duality and finite comparisons

The current-to-height correspondence below is the standard planar XY duality; see [14, Section 2] and [15]. We give the finite free-box identity to fix its boundary and winding conventions.

Lemma 2.1 (Free-boundary duality). *In a finite free spin rectangle, the two-point spin correlation is the ratio of a twisted height partition sum to the centered height partition sum, both with weights p_b and exterior face height zero. The twist is an integral connection with circulations $+2\pi, -2\pi$ around the two source vertices and zero circulation elsewhere. For interior sources it can vanish near the outer boundary. After cuts, the same description holds on each retained chart, without identifying heights through the cut.*

Proof. Expand each factor $e^{b \cos(\theta_x - \theta_y)}$ in its Fourier series. Integrating the spins imposes zero divergence on the integer edge current in the denominator. Inserting $e^{i(\theta_0 - \theta_x)}$ changes its divergence to the two unit sources. The insertion with the opposite sign has the same sum by current reversal, so taking the real part makes no change. The factors e^b cancel between the two partition sums.

Rotate currents onto dual edges. In a planar rectangle, every divergence-free integral current is a difference of integer face heights with exterior height zero. This follows by summing the rotated current along dual paths: path independence is exactly the zero-divergence condition at the enclosed primal vertices. Subtract a fixed unit current on a path between the sources in the numerator. The remaining current is divergence free and gives the same height variables. The subtracted current is the claimed connection. Changing its path is an integral change of height variables. Equivalently, the numerator has multivalued heights with the stated monodromies and single-valued branches on simply connected charts. Releasing identifications across a cut gives the asserted cut version. $\square$

The finite Gaussian inequalities of [12, Lemma 2.1], based on the lattice inequality of Regev and Stephens-Davidowitz [13], apply to this nonquadratic weight through a series approximation. We record the extension, including the forms needed after conditioning. The general use of Gaussian representations for this transfer has precedent in [1, Lemmas 9.6–9.7 and Appendices C–D]; the Bessel mixture representation there is credited to Aran Raoufi. The series construction below verifies the particular affine constraints and observation forms used here. It does not require the bare Bessel potential to satisfy a super-Gaussian condition at every value of b ; subdivision is used in such extensions, as also in [9, Theorem 8 and Section 1.3.3].

Lemma 2.2 (Finite comparison rules). *On a fixed primary graph, allow additional nonnegative quadratic forms in real linear observations, centered equalities, and independent Gaussian observation noises.*

- (i) *Increasing centered precision decreases centered covariance and every centered real moment generating function. Covariance on a translate, including after a real linear tilt, dominates centered covariance. In particular*

$$\log \mathbb{E} e^X \geq \frac{1}{2} \text{Var}(X)$$

for a centered inversion-symmetric law and a real linear test X .

- (ii) *A shifted-to-centered partition ratio is at most one and decreases when the same positive precision is added to the affine and centered coordinates. Shifts may be nonzero equality values, an integral connection, or centers of observation penalties. The monotonicity also holds with a fixed real linear statistic in the numerator exponent, but the bound by one need not hold for this tilted ratio. Equality constraints may be inserted by infinite penalties whenever the affine constraint is nonempty.*
- (iii) *Centered characteristic coefficients are nonnegative and increase with precision. Their logarithms have Hessian at every phase at least the Hessian at zero. A phase on a free component is allowed if its coefficient total is an integer. In the invariant finite-graph situations used below these coefficients are positive.*

Released constant modes are handled modulo translations or by confining observations. Real tests used when releasing a constant annihilate that constant.

Proof. Replace every primary edge by N bonds in series, each with normalized integer increment weight

$$q_N^{j^2}, \quad q_N = b/(2N).$$

For all sufficiently large N this is a genuine lattice Gaussian. For every fixed real t , uniformly on compact sets of t ,

$$\frac{\sum_j q_N^{j^2} e^{tj}}{\sum_j q_N^{j^2}} = 1 + \frac{b}{N} (\cosh t - 1) + O_t(N^{-2}).$$

The sum of the N increments therefore converges to p_b , with convergence of exponential moments at every fixed real argument. Matching chain endpoints and imposing the cycle and period constraints gives a quadratic lattice law on the enlarged graph. The numerator is an affine lattice parallel to the denominator lattice.

All partition ratios just listed converge at fixed primary graph. For domination, use a spanning forest to express vertex differences as sums of chain increments; the remaining normalized edge weights are bounded by one. The forest variables have uniformly bounded exponential moments. If

a constant is confined, sum its translations against the confining Gaussian. This also dominates fixed real tilts and polynomial factors. Normalizers of the independent chain laws agree in the two sectors.

On the enlarged lattice, the covariance and partition comparisons are precisely [12, Lemma 2.1]. They hold on a proper affine subspace by restricting to its parallel span and completing the square in the remaining fixed coordinates. For clarity, with quadratic precision $T + tQ$, $Q \geq 0$, differentiation of the log shifted-to-centered ratio gives

$$-\frac{1}{2} \operatorname{tr} Q (\operatorname{Cov}_{\text{shift}} - \operatorname{Cov}_0 + m_{\text{shift}} \otimes m_{\text{shift}}) \leq 0.$$

After a fixed real tilt, completing the square gives the same formula with the tilted covariance and mean. This proves the tilted version of (ii), not merely its untilted version. Infinite centered penalties implement the required equalities. Poisson summation on the enlarged lattice proves (iii). Taking the fixed-graph limit proves the stated rules. Vanishing auxiliary masses justify released translation modes; alternatively one can use integer gradient coordinates. $\square$

The bounds that remain uniform in subdivision are not the bare Gaussian bounds of the enlarged network. They are bounds on whole chain increments.

Lemma 2.3 (Reference-cost bounds). *Suppose a primary test has a flow representation*

$$X = \sum_e w_e \nabla_e h, \quad \mathcal{R}(w) = \sum_e w_e^2.$$

In a centered law with any additional centered precision,

$$\operatorname{Var}(X) \leq C_b \mathcal{R}(w), \quad \mathbb{E}|X|^r \leq C_{r,b} \mathcal{R}(w)^{r/2}. \quad (2.1)$$

For each fixed t , $\mathbb{E} \exp(tX/\sqrt{\mathcal{R}(w)})$ is bounded uniformly in the graph and in sufficiently fine subdivision. The constants can be uniform for b in a compact subset of $(0, \infty)$. These statements include minimization over flows with the prescribed divergence and grounded vertices.

Proof. Keep independent full-chain increments as variables and release their endpoint, curl, and period constraints. Lemma 2.2 bounds the original centered transform by this product transform. For one subdivision bond and bounded t , pairing opposite increments in its moment generating function gives

$$\log \frac{\sum_j q_N^{j^2} e^{2\pi t j}}{\sum_j q_N^{j^2}} \leq C_{b,R} t^2 / N \quad (|t| \leq R).$$

Summing over N bonds gives $C_{b,R} t^2$ for a whole chain. For a flow normalized by $\sqrt{\mathcal{R}(w)}$, every coefficient has absolute value at most one. Independence therefore bounds its log transform at a fixed real argument by a constant. This proves the normalized exponential and moment bounds, and rescaling proves (2.1). The same inequalities on the original law follow by the fixed-graph limit. In this argument a centered primary observation is a real linear function of the enlarged coordinates, so its squared penalty is an additional positive form. Cycle, period, and endpoint conditions are likewise limits of positive equality penalties. They can all be removed before applying the independent whole-chain bound. No uniform estimate on the inverse microscopic Gaussian precision is asserted or used. $\square$

We call $\mathcal{R}$, or its minimum over flows, the *reference cost*. In particular it always refers to the primary unit-edge graph. For a zero-total profile f , this minimum is $\langle f, A^+ f \rangle$, by the usual orthogonal decomposition of flows into a minimizing gradient flow and a divergence-free flow.

There are two useful further consequences. First, if a phase coefficient $\phi(s) > 0$ is varied by a zero-total test Y of variance at most V , the logarithmic Hessian bound gives

$$\sqrt{-\log \phi(s + tY)} \leq \sqrt{-\log \phi(s)} + |t| \sqrt{V/2}. \quad (2.2)$$

Indeed $f = \log \phi \leq 0$, $f'' \geq -V$, imply $|f'|^2 \leq -2Vf$, which integrates to the displayed inequality. This proves the same transfer bound as [12, Equation (2.6)], now with centered variance and hence reference cost. An integer-total phase on a fixed free graph is equivalent to a zero-total phase after subtracting an integer point charge. Equation (2.2) starting at the zero phase proves its strict positivity.

Second, centered restoration comparisons can be localized. Represent a removed edge by an independent chain and restore it by the centered equality matching its endpoint difference. Adding exterior bonds or centered guard pins increases the probability of these equalities. The centered restoration ratio is therefore between its value in a free patch and its value with a separating guard pinned to zero. At that pin the interior no longer depends on the exterior, provided no observation crosses the guard. If all endpoints of the restored bonds, and the support of any restored form, lie in H , pinning every primary vertex of H makes restoration a fixed scalar, also for a chain. Bayes' rule and centered precision comparison bound the free-to-guard multiplicative gap by

$$\frac{\mathbb{P}(h_p = 0 \mid h_H = 0)}{\mathbb{P}(h_p = 0)}. \quad (2.3)$$

These assertions also follow by interpolating positive forms before taking equality limits. Independent-chain scalar normalizers are the same in each member of the comparison.

2.2 Free Gaussian limits

For $B_m = \{0, \dots, m-1\}^2$, put $g_m(x) = x_1/m$ and

$$a_m(b) = \text{Var}_{B_m, b} \langle h, A_{B_m} g_m \rangle.$$

No bare quadratic parameter is attached to this definition.

Theorem 2.4 (The free coefficient). *There exists a finite $a = a(b) \geq 0$ such that $a_m(b) \rightarrow a$. Let B_n , at mesh n^{-1} , be any sequence of free axis rectangles with endpoints converging to those of a nondegenerate rectangle B . For every finite collection of smooth shifts defined near $\overline{B}$,*

$$\log \mathbb{E} \exp \{ \langle h, A_{B_n} g(\cdot/n) \rangle \} \longrightarrow \frac{a}{2} \int_B |\nabla g|^2. \quad (2.4)$$

Joint limits are Gaussian, with covariance $a \int_B \nabla g \cdot \nabla g'$, and all joint moments converge. For zero-total profiles f_n whose scaled densities converge in $L^2(B)$ to a bounded piecewise continuous f of integral zero,

$$\log \mathbb{E} e^{\langle h, f_n \rangle} \longrightarrow \frac{a}{2} \langle f, (-\Delta_{B, N})^{-1} f \rangle_{L^2(B)}. \quad (2.5)$$

Face smears, after subtracting their totals from a volume smear, are allowed as well. These limits may be degenerate when $a = 0$.

Proof. We give the changes to the cutting and folding proof of [12, Theorem 2.2]; the observation Laplacian here is the primary A , not the subdivided quadratic precision.

Cut an M -square into m -squares and strips of width less than m . In its affine observation, the seam and strip flows have reference cost $O(m^{-1} + m/M)$. Cutting increases covariance, and the retained cell observations become independent. Consequently

$$\sqrt{a_M} \leq \sqrt{a_m} + C_b(m^{-1} + m/M)^{1/2}. \quad (2.6)$$

First let $M \rightarrow \infty$, and then let m follow a liminf subsequence. This proves existence and finiteness of a . Square symmetry makes the limiting covariance of the two affine observations scalar.

Cutting rectangles into squares gives the upper variance bound $a \int_B |\nabla g|^2$ for affine g . For the reverse bound, at each fixed discrete rectangle tile an arbitrarily large square by copies of that rectangle, choosing its side divisible by both rectangle side counts. Apply the same seam estimate, first sending that square side to infinity. If $m_1(n), m_2(n)$ are the rectangle side counts and $V_n(p)$ is its variance for $g = p \cdot x/n$, this gives

$$\sqrt{a} |p| \leq \frac{n}{\sqrt{m_1(n)m_2(n)}} \sqrt{V_n(p)} + \frac{C_{b,p}}{\sqrt{\min(m_1(n), m_2(n))}}.$$

Thus every affine shift attains the upper bound along every admissible rectangle sequence.

Cut into cells of small fixed macroscopic diameter and approximate each gradient there by a constant. Seams have vanishing reference cost, and the interior error is controlled by the squared L^2 -modulus of continuity. This proves

$$\limsup_n \text{Var}\langle h, A_{B_n} g \rangle \leq a \int_B |\nabla g|^2 \quad (2.7)$$

for smooth or continuous piecewise smooth g , and for partial gradient observations restricted to polygonal pieces. Call equality in this upper bound *saturation*. Saturated shifts form a linear space: on a convergent covariance subsequence the upper-bound defect is positive semidefinite, and its diagonal vanishes for saturated shifts, so every entry vanishes. Saturation passes to an enclosed rectangle by cutting its complement into rectangles and comparing the sum of their upper bounds with the saturated outer variance. It also passes through gradient-energy approximation, by Lemma 2.3.

To obtain nonaffine saturated shifts, condition on the whole trace of a site-axis or diagonal reflection line and pin one of its vertices. The subdivided quadratic graph is attractive: its density satisfies the lattice condition edge by edge. Its conditioned law is associated, and association passes to the fixed-primary-graph limit. If a saturated seed is odd about the line, the two partial tests on its open sides are reflected negatives. If the off-line coefficients on one side are nonnegative after a vanishing-cost transport to the line, conditional association gives nonpositive conditional cross covariance. The two conditional means are negatives, so the unconditional cross covariance is nonpositive as well, up to the vanishing transport error. The saturated sum and the separate upper bounds (2.7) force each partial test to saturate. The corresponding folded shift differs from that partial test only along a vanishing-cost seam.

Here the coefficient verification can be made on unit edges. For the axis seed $g = x_1$, the only negative coefficients of the right-side partial observation occur next to the cut. Move them to the conditioned line along bounded-length paths. There are $O(n)$ coefficients of size $O(n^{-1})$, with bounded overlap, so the cost is $O(n^{-1})$. The hinges $(x_1 - t)_+$, and then all smooth univariate functions, are therefore saturated. For a diagonal seed, put $Y = y + t$ and take $g = F(x) - F(Y)$ on a symmetric enclosing square $p \leq x, Y \leq q$. Initially assume $F''' < 0$ and $F'(p), F'(q) > 0$. In

$x > Y$, away from the diagonal strip, the interior coefficient is nonnegative because the symmetric second differences of F decrease with their center. The coefficients on $x = q$ and $Y = p$ have positive leading terms $F'(q)/n$ and $F'(p)/n$; their $O(n^{-2})$ remainders are harmless. Move the exceptional diagonal-strip coefficients to the line. Their total path cost is again $O(n^{-1})$. The fold criterion proves saturation of

$$(F(x) - F(y + t)) 1_{\{x > y+t\}}. \quad (2.8)$$

Every smooth F is a difference of two functions with the strict properties just used: add a sufficiently large cubic with negative third derivative and then a sufficiently large positive linear function. Hence (2.8) holds for arbitrary F . Coherently round a reflection line by $O(n^{-1})$. The changed fold has squared gradient cost $O(n^{-1})$: only an $O(n^{-1})$ -width strip changes its side, and away from that strip the smooth gradient changes by $O(n^{-1})$. Restriction from the enclosing symmetric rectangle consequently gives the claim on every admissible target rectangle sequence.

Finally choose a smooth family F_t , compactly supported in the relevant t -range, with $F'_t(x) = -g_{xy}(x, x - t)$. The mixed derivative of the integral of (2.8) is g_{xy} . Its difference from g is therefore a sum of univariate functions. Riemann sums converge in gradient L^2 , including across the moving diagonals. Linearity and energy approximation prove saturation for every smooth g .

The lower Laplace bound now follows from Lemma 2.2. For the upper bound, cut into cells of side comparable to a fixed r . Remove seam errors by Holder's inequality and the reference-cost exponential bound. A cell test has reference-cost square root $O(r)$; hence, for fixed $p > 1$,

$$\log \mathbb{E} e^{pX} = \frac{1}{2} p^2 \text{Var } X + O_p(r^3).$$

The remainder follows from $\mathbb{E}[|X|^3 e^{p|X|}] = O_p(r^3)$, which uses normalized exponential moments, not a bare Gaussian upper tail. There are $O(r^{-2})$ cells. First take the microscopic limit, then $r \downarrow 0$, then $p \downarrow 1$. This proves (2.4); applying it to real linear combinations proves the joint assertion and moment convergence.

Nearest-neighbor Poincare bounds the reference cost of a density error by a constant times its scaled L^2 -norm squared. Approximate f by finite sums of Neumann cosines, use their smooth inverse-Laplacian shifts, and then remove the approximation. For such a shift $n^2 A_{B_n} g$ converges in density L^2 to $-\Delta g$; the boundary coefficients are $O(n^{-2})$ and occupy $O(n)$ sites. Using the actual rectangle endpoints before letting them converge handles arbitrary admissible sequences. This proves (2.5). A face probability can be moved to a strip of width r at reference cost $O(r + n^{-1})$, by parallel normal flows. Strip approximation proves the face extension. $\square$

2.3 Soft joining and the coefficient gap

Write u_m for uniform probability on B_m , and

$$w_m(s) = \mathbb{E}_{B_m, b} e^{i\langle h, s \rangle}, \quad \sum_x s_x \in \mathbb{Z}.$$

The definition is independent of the representative modulo $2\pi\mathbb{Z}$, and $0 < w_m(s) \leq 1$.

Theorem 2.5 (Uniform phase and forbidden range). *For every $b > 0$,*

$$a(b) = 0 \quad \text{or} \quad a(b) \geq 8\pi.$$

If $a(b) > 0$, the fractional mean $\langle h, u_m \rangle \bmod 2\pi$ converges to uniform measure on the circle, independently of any fixed finite collection of the Gaussian observations of Theorem 2.4. More generally, for a fixed nonzero integer l and any zero-total profiles t_m of bounded reference cost,

$$w_m(lu_m + t_m) \longrightarrow 0. \quad (2.9)$$

Proof. We first isolate the replacement for the quadratic charge-space joining used in [12, Theorem 2.3] and [12, Lemma 2.4]. Tile a rectangle by free cells, keeping their lattice translations, and independently assign every seam edge a difference of law $2\pi p_b$. Let $\mathcal{V}$ be the neutral Euclidean space of cell totals. Choose a linear flow right inverse P on the neighboring-cell graph, so $\text{div } P(d) = d$ for $d \in \mathcal{V}$. Define $H \in \mathcal{V}$ by

$$\langle H, d \rangle = \sum_{\langle i, j \rangle} P(d)_{ij} \left(\langle h_i, \rho_{ij} \rangle - \langle h_j, \rho_{ji} \rangle - \bar{\eta}_{ij} \right). \quad (2.10)$$

Here ρ_{ij} is the uniform seam-face probability and $\bar{\eta}_{ij}$ is the mean of the independent seam differences, with consistent orientation. Translating the cells by constants c_i changes H by the orthogonal projection of c onto $\mathcal{V}$. Weight this law by $e^{-|H|^2/2}$, summing relative translations and taking the common translation modulo 2π . Adding exact seam matching makes $H = 0$ and recovers the joined rectangle. The soft law thus has larger internal variances and smaller invariant characteristic coefficients than the joined law. These comparisons with translation-summed laws can be justified without biasing the initial free-cell laws. First add the auxiliary precision $\eta \sum_i \bar{h}_i^2$, where $\bar{h}_i$ is the cell mean. For a fixed cell shape with representative mean μ , its translation sum is $\sum_{k \in \mathbb{Z}} e^{-\eta(\mu+2\pi k)^2/2}$. After division by its Gaussian integral scalar, this tends uniformly in μ to one and is uniformly bounded, by Poisson summation. Thus the original internal law tends to the product of the free-cell laws, also in moments of invariant tests. After adding $|H|^2$, divide out the common translation scalar. The relative translation sum is a periodized Gaussian on the projected lattice, uniformly bounded in its shape-dependent shift. Dominated convergence again applies with the internal moments. Apply precision comparison at $\eta > 0$ and then let $\eta \downarrow 0$. In particular the soft variance of the sum of two internal affine tests is at most their independent free-cell variance.

For a phase whose cell restrictions are s_i , of totals ξ_i , Poisson summation in the relative translations gives, up to a common scalar, the numerator

$$\sum_{\substack{n_i \in \mathbb{Z} \\ \sum n_i = \sum \xi_i}} e^{-|d|^2/2} \prod_i w_i \left(s_i + \sum_{j \sim i} P(d)_{ij} \rho_{ij} \right) \chi_m(P(d)), \quad d = n - \xi. \quad (2.11)$$

The denominator uses $s = 0$. The signs of P are chosen so the displayed cell totals are n_i . Indeed the reciprocal vectors of the projected translation lattice are exactly the neutral integer vectors; the phase shifts them to $n - \xi$. Fourier transforming the Gaussian soft weight supplies $e^{-|d|^2/2}$; the remaining factors are the cell and independent-seam characteristic functions. The latter factor χ_m is positive and at most one. For fixed cell fluxes it tends to one, since each seam average has variance $O(m^{-1})$. For two cells one may use a scalar mismatch penalty, giving the same formula with integer sectors l and weights $e^{-l^2/2}$. All these calculations can first be performed on the subdivided quadratic graph and then passed to the fixed primary graph.

We prove the uniform phase. Let ρ be the right-face probability, and set

$$X = \langle h, A(x_1/m) \rangle, \quad Y = \langle h, \rho - u_m \rangle.$$

On the unit square graph X is exactly the right-face average minus the left-face average. Theorem 2.4 gives joint Gaussian limits, $\text{Var } X \rightarrow a$, and, by reflection, $\text{Cov}(X, Y) \rightarrow a/2$. Suppose, for some nonzero integer l , that $w_m(lu_m)$ stays bounded away from zero on a subsequence. The reference-cost exponential bounds give a further subsequence on which

$$F_m(t, v) = \mathbb{E} e^{il \langle h, u_m \rangle + itX + ivY}$$

converges locally with all derivatives to an entire function F . On real arguments it is positive: this follows from (2.2) and the assumed positive value at the origin.

Here is the replica calculation responsible for its form. For two independent copies of a centered subdivided quadratic height, put $S = h_1 + h_2$, $D = h_1 - h_2$. Conditioned on their common parity class modulo $4\pi\mathbb{Z}$, they are independent and identically distributed. Writing $V_{ij} = \mathbb{E}(S_i S_j \mid \text{parity})$, we have

$$\mathbb{E}V_{ij} = 2 \text{Cov}(h_i, h_j), \quad \text{Cov}(V_{ij}, V_{rs}) = 2\kappa(h_i, h_j, h_r, h_s). \quad (2.12)$$

To verify the second identity, put $C_{ij} = \mathbb{E}h_i h_j$ and $M_{ijrs} = \mathbb{E}h_i h_j h_r h_s$. Conditional independence and expansion in the two original copies give

$$\mathbb{E}V_{ij}V_{rs} = \mathbb{E}S_i S_j D_r D_s = 2M_{ijrs} + 2C_{ij}C_{rs} - 2C_{ir}C_{js} - 2C_{is}C_{jr}.$$

Subtracting $\mathbb{E}V_{ij}\mathbb{E}V_{rs} = 4C_{ij}C_{rs}$ gives exactly twice the fourth joint cumulant, without symmetrizing the indices. In particular $\text{Var}(V_{XX}) = 2\kappa(X, X, X, X)$ and $\text{Var}(V_{XY}) = 2\kappa(X, X, Y, Y)$. After first taking the fixed-graph chain limit and then the Gaussian limit, both vanish.

For completeness, write $F_{m,N}$ for the transform before the chain limit, and put $\theta(h) = l\langle h, u_m \rangle + tX + vY$. Inversion symmetry makes $F_{m,N}$ real on real arguments. For the observation indices $i, j \in \{X, Y\}$, differentiating in their respective phase variables gives

$$\mathbb{E}S_i S_j e^{i\theta(D)} = -2(F_{m,N}F_{m,N,ij} - F_{m,N,i}F_{m,N,j}).$$

By conditional independence the left side equals $\mathbb{E}[V_{ij}\mathbb{E}(e^{i\theta(D)} \mid \text{parity})]$. Replacing V_{ij} by $2C_{ij}$ has absolute error at most $\sqrt{\text{Var}(V_{ij})}$, and $\mathbb{E}e^{i\theta(D)} = F_{m,N}^2$. Fixed-graph moment convergence passes this bound through $N \rightarrow \infty$; Gaussian moment convergence then removes the error as $m \rightarrow \infty$. Thus

$$-2FF_{tt} + 2F_t^2 = 2aF^2, \quad -2FF_{tv} + 2F_tF_v = aF^2.$$

Since $F > 0$ on real arguments, division by $-2F^2$ yields

$$(\log F)_{tt} = -a, \quad (\log F)_{tv} = -a/2.$$

Reflection gives $F_t(0, 0) = 0$; hence

$$F(t, v) = F(0, v) \exp(-at^2/2 - atv/2). \quad (2.13)$$

Now softly join two horizontal m -squares and test the sum of their internal affine observations. Its variance is at most twice the free-cell variance, and at least its variance in the fully joined rectangle. The joined rectangle's full affine test differs from this sum by the seam test, whose reference cost is $O(m^{-1})$: there are m seam coefficients of size m^{-1} . Its limiting variance is consequently $2a$, so the soft-law variance also tends to $2a$.

In (2.11), every sector function has logarithmic second derivative at least minus twice the free-cell variance. The logarithmic second derivative of a positive sum is the weighted mean of these derivatives plus the variance of the sector logarithmic first derivatives. Thus the negative second derivative of the normalized sum, the soft-law variance, is at most twice the free variance minus that nonnegative sector variance. Differentiation of the sector sum is justified by the following summable bounds. Write $\psi_l(t)$ for the product of the two cell characteristic coefficients in sector l , and $V = 2 \text{Var}(X)$ for the independent-cell variance. Then $0 < \psi_l \leq 1$ and $(\log \psi_l)'' \geq -V$, so the derivative estimate underlying (2.2) gives

$$\psi_l |(\log \psi_l)'|^2 \leq 2V(-\psi_l \log \psi_l) \leq 2V/e.$$

Multiplication by $e^{-l^2/2}\chi_m(l) \leq e^{-l^2/2}$ gives a summable bound, uniform in m , on the logarithmic-derivative square terms. The raw second derivatives are bounded by the independent-cell second moment V . The analogous first-derivative bound follows by Cauchy–Schwarz. Hence termwise differentiation and the mixture-Hessian identity are valid even for sectors whose coefficients become small. The sector zero has weight one before normalization. The sector l stays bounded away from zero: transfer from lu_m to the seam face costs a bounded amount, and $\chi_m \rightarrow 1$. By (2.13) at $v = l$, reflection and charge reversal give its logarithmic first derivative tending to $-al$, up to the irrelevant common orientation. The zero sector derivative is zero. If $a > 0$, this contradicts the vanishing sector variance. Therefore $w_m(lu_m) \rightarrow 0$. Equation (2.2) proves (2.9); Fourier approximation on the circle and the Gaussian limits prove the joint uniform-phase assertion.

It remains to obtain the sharp numerical gap. Let $\bar{\rho}_m$ be one quarter of the sum of the four face probabilities and put $z_m = w_m(\bar{\rho}_m) > 0$. Tile a Gm -square into G^2 cells, with G fixed. In the outer-face phase, the cell totals ξ_i are supported at the boundary, sum to one, and satisfy $\sum_i \xi_i^2 \leq C/G$. The denominator of (2.11) tends to one: its zero sector is one; every other sector has a cell of nonzero integer total and tends to zero by (2.9); Gaussian sector weights permit dominated convergence at fixed G .

Retain numerator sectors $n = e_{i_0}$, where i_0 is at least $G/4$ from the boundary. The radial-flow construction of [12, Equation (2.25)] supplies outward face fluxes H_{ie} with divergence e_{i_0} , constant outer flux $1/(4G)$, and, on putting

$$D_{ij} = (H_{i,+j} - H_{i,-j})/2, \quad S_{ij} = H_{i,+j} + H_{i,-j},$$

the bounds

$$\sum_{i,j} D_{ij}^2 \leq \frac{1}{2\pi} \log G + C, \quad \sum_{i,j} S_{ij}^2 \leq C. \quad (2.14)$$

For clarity, its continuum field is

$$V(x) = \frac{x - x_0}{2\pi|x - x_0|^2} + E(x), \quad |E| \leq C/G, \quad |\nabla E| \leq C/G^2.$$

The correction fixes the constant outer normal flux; it is obtained by the zero-Neumann problem after even reflection. Integrating V across unit-cell faces gives H . The squared radial field integrates to $(2\pi)^{-1} \log G + O(1)$; face-sum derivative errors are summable and give the second bound in (2.14). These constants are uniform for the retained central cells. The internal flows have divergences $e_{i_0} - \xi$. These vectors are linearly independent, since their central coordinates are distinct and ξ is supported on boundary cells. Prescribe P on them by these flows and extend linearly. Thus one soft precision works for all retained sectors.

In each cell, expand the logarithm of its characteristic coefficient around the symmetric four-face profile of the same total, either zero or $\bar{\rho}_m$. The linear term vanishes by square symmetry. The logarithmic Hessian lower bound therefore gives a product lower bound

$$z_m \exp \left\{ -\frac{1}{2} \sum_i \text{Var} \langle h, Y_i \rangle \right\},$$

where Y_i is the face-profile difference. Coordinate reflections split Y_i into two mutually orthogonal odd face differences and an even remainder. The odd differences have limiting variance a ; the even remainder has reference cost bounded by $C \sum_j S_{ij}^2$. Consequently, as also calculated in [12, Equation (2.23)],

$$\limsup_m \sum_i \text{Var} \langle h, Y_i \rangle \leq a \sum_{i,j} D_{ij}^2 + C_b \sum_{i,j} S_{ij}^2.$$

The soft penalty is bounded below uniformly on these sectors, since $|e_{i_0} - \xi|^2$ is bounded, and the seam factor tends to one at fixed G . There are at least cG^2 retained sectors. Characteristic monotonicity from the soft law to the joined law yields

$$\liminf_{m \rightarrow \infty} \frac{z_{Gm}}{z_m} \geq c_b G^{2-a/(4\pi)}, \quad (2.15)$$

with $c_b > 0$ independent of sufficiently large fixed G . If $0 < a < 8\pi$, choose one integer G making the right side larger than two. Then $z_{Gm} \geq (3/2)z_m$ for every sufficiently large m . Iterating along $m, Gm, G^2m, \dots$ contradicts $0 < z_m \leq 1$. This proves the gap. $\square$

2.4 An upper bound for spin correlations

Proposition 2.6 (Spin upper exponent). *For the prescribed free-box correlation,*

$$\limsup_{n \rightarrow \infty} \frac{\log C_b(n)}{\log n} \leq -\frac{2\pi}{a(b)}. \quad (2.16)$$

When $a(b) = 0$, the assertion means that the limsup is $-\infty$, or equivalently that every fixed polynomial upper exponent is eventually available.

Proof. In a sufficiently large free spin box, put disjoint circular dyadic annuli of radii $r, 2r$ around each source, through radii of order n . On the fixed annulus $1 < |z| < 2$, choose a smooth tangential divergence-free form p , compactly supported there and approximating $d \arg z$ in L^2 . A radial cutoff provides such forms. Discretize p at scale r by taking differences of a sampled stream function on faces. Its pairing with twisted differences is deterministic, by circulation, and tends to

$$D = \int p \cdot \nabla \arg z.$$

Use the opposite sign at the negative source.

Insert in the twisted numerator the real exponential of λ times the sum of these deterministic pairings. Release endpoint constraints between a grid of small rectangular cells at every annular scale, retaining independent seam increments; release all remaining exterior constraints as well. Lemma 2.2(ii), with its real tilt, bounds this normalized tilted numerator by the resulting product of transforms. Each retained cell is simply connected and avoids the sources, so its integral connection disappears by an integer gauge change. Here the tilt is on the affine edge differences, the same coordinates as the edge weights. If the connection on a chart is $\alpha = \nabla \kappa$, the reindexing $h' = h + \kappa$ replaces $\nabla h + \alpha$ by $\nabla h'$ in both the weight and the tilt. It therefore introduces no residual real-tilt scalar.

Take cells of side comparable to tr , lying within the annulus, with $t > 0$ small and fixed. There are only $O_t(r)$ leftover nonzero edge coefficients, each $O(r^{-1})$; their independent-chain log moment is $o_r(1)$. On the squares, approximate the coefficients by constant gradients. Theorem 2.4, Holder's inequality, and the reference-cost bound on the coefficient error give log transform at most

$$\frac{\lambda^2 a}{2} \int |p|^2 + o_r(1) + o_t(1).$$

The errors are taken first at fixed t, λ, p ; rounding changes only vanishing reference-cost terms.

There are $\log n / \log 2 + O(1)$ annuli per source. At any fixed n all cuts are internal to a sufficiently large box, so we may first take its prescribed free-box limit. Averaging the convergent estimates along the increasing annular scales gives

$$\limsup_n \frac{\log C_b(n)}{\log n} \leq \frac{2}{\log 2} \left(-\lambda D + \frac{\lambda^2 a}{2} \int |p|^2 \right),$$

after $t \downarrow 0$. Now let $p \rightarrow d \arg z$; both D and $\int |p|^2$ tend to $2\pi \log 2$. For $a > 0$, the optimal choice $\lambda = 1/a$ yields $-2\pi/a$. The factor two is from the two sources. For $a = 0$, each fixed $\lambda > 0$ is allowed, giving $-4\pi\lambda$; letting this fixed parameter be arbitrarily large proves the stated interpretation. $\square$

3 Bands, primary hard pins, and conditional data

Fix $b > 0$ with $a = a(b) > 0$. The free limits from Section 2 must be transferred to hard pins, and then to conditional laws at random guard data. These are stronger statements than convergence of finitely many unconditioned smears. We prove the transfer using the cable exploration of [12, Lemma 3.2] and [12, Proposition 3.7], with primary reference costs replacing bare Gaussian variances. The loss of Gaussian tails under subdivision is handled by polynomial damping and a separate truncation of large tilts.

3.1 Geometry and the required limits

Throughout this section a fixed geometry is a finite union of bounded rectangular pieces with nondegenerate faces and widths. Pieces in one component are joined along full interfaces, which may be subdivided into segments. Point contacts do not identify components, and opposite sectors are not joined only through a vertex. Equivalently, we use the polygonal Lipschitz geometries of [12, Definition 3.11], including the cut geometries there. The nearest-neighbor primary lattice has mesh n^{-1} . Aligned boundaries may be rounded by a bounded number of lattice steps, preserving the faces, corners, and positive-width passages. All assertions also hold along sequences of such roundings. A hard pin on a region means that every *primary* height in that region is zero; it does not pin the internal subdivision vertices. Pin regions are aligned unions of positive-area boxes.

The ordinary unit-edge real Dirichlet form is the reference form. On a component U with pin region p , its limiting energy space is

$$\mathcal{H}_{U,p} = \{v \in H^1(U) : v = 0 \text{ almost everywhere on } p\}, \quad \mathcal{E}_U(v) = \int_U |\nabla v|^2.$$

For an empty pin we take this space modulo constants when needed. Free walls are understood in the variational, hence Neumann, sense. The real form approximation, Poincare estimates, and energy density statements in [12, Lemma 3.12] apply to these ordinary nearest-neighbor forms.

Theorem 3.1 (Mixed height limits). *For every fixed geometry above and every $a(b) > 0$, the massless height law with primary hard pins has Gaussian Laplace and moment limits for fixed bounded, piecewise continuous density smears. On a component meeting a pin, the covariance is the inverse of*

$$a^{-1} \mathcal{E}_U \quad \text{on } \mathcal{H}_{U,p}.$$

On an unpinned component this assertion concerns mean-free smears. Its fractional mean modulo 2π is asymptotically uniform and independent of the mean-free Gaussian field.

Adding a fixed finite-rank nonnegative quadratic penalty in box averages gives the corresponding Gaussian law. If the penalty controls every unpinned component constant, lattice translations are summed and the limiting constants are integrated; the result is an ordinary, unprojected Gaussian law. The assertions include energy-approximable shifts and hold along every admissible primary mesh sequence.

The second conclusion concerns the centered *high law* μ_n , obtained by adding $\|B_n h\|^2/2$, where B_n is a fixed finite collection of box averages controlling all component constants. Each defining box

lies wholly in one pre-pin component, so this high-law penalty is a sum of component-local terms. For positively separated box regions H, p , initially without hard pins, put

$$D_n(H, p) = \log \frac{\mu_n(h_H = 0 \mid h_p = 0)}{\mu_n(h_H = 0)}, \quad r_{H,n}(b^p) = \frac{\mu_n(h_H = 0 \mid h_p = b^p)}{\mu_n(h_H = 0 \mid h_p = 0)}. \quad (3.1)$$

Regions lying in different components contribute independently.

We use the following precise meaning of convergence at random pin data. Choose an increasing total list of smooth density probes inside p . A function of h_p has a Gaussian *finite-observation approximation* if, for every tolerance, it can be approximated in probability by a function of a sufficiently long fixed list of these probes, whose joint law and function converge to the Gaussian counterparts. The probe list is increased after the primary lattice limit. No pointwise continuum trace, or uniform approximation at all microscopic pin values, is asserted.

Theorem 3.2 (Separated pin interaction and conditional laws). *Under the centered high law, $D_n(H, p) \geq 0$ converges to the Gaussian pin interaction for precision*

$$a^{-1} \mathcal{E}_U(v) + \|Bv\|^2.$$

For every fixed finite list of density or box smears X , the conditional Laplace transforms and conditional weak laws given h_p have the Gaussian finite-observation approximations just defined. Moreover,

$$0 < r_{H,n} \leq 1, \quad \mathbb{E}_{\mu_n} r_{H,n} = e^{-D_n(H,p)},$$

and $r_{H,n}$ has the corresponding Gaussian approximation, whose limit is strictly positive almost surely. The convergence assertions persist when data are sampled under a fixed finite linear exponential tilt, with D_n and $r_{H,n}$ retaining their centered definitions and the limiting law sampled under the corresponding Gaussian tilt. The displayed expectation identity concerns μ_n , not its tilts. Conditional transforms for a tilted baseline also converge, by taking ratios of the centered conditional transforms.

Both theorems are proved below. The Gaussian determinant and localization facts used in the second theorem are [12, Proposition 3.14], applied to the ordinary real nearest-neighbor form, not to the subdivided bare form.

3.2 A confining band and uniform primary estimates

It is enough initially to work in one fixed free component U . Choose a positive-width band K between a starting region and a test region, with positive gaps on both sides. Partition K into rectangles d of side comparable to ℓ and bounded aspect ratio. Write h_d for a primary probability average and $|d|$ for its scaled counting area. Define

$$W_{\ell,\lambda}(h) = \exp \left\{ -\frac{\lambda}{2} \sum_d |d| h_d^2 \right\}. \quad (3.2)$$

Let $\mathcal{P}_n$ be the sigma-finite free measure obtained by normalizing one representative of each common-translation orbit and summing $h + 2\pi m$, $m \in \mathbb{Z}$. Changing the representative only relabels this sum. The band law is

$$d\mathcal{P}_{n,W} = \widehat{W} d\mathcal{P}_n, \quad \widehat{W} = W/\mathcal{P}_n(W).$$

Its denominator is finite and positive. We always take limits in the order

$$n \longrightarrow \infty, \quad \ell \downarrow 0, \quad \lambda \longrightarrow \infty. \quad (3.3)$$

Subdivision size N is taken sufficiently large at each fixed primary graph, or first through a fixed-graph limsup in the uniform estimates below.

Before introducing pins, the free Gaussian limits on U follow from Theorem 2.4 by the enclosing-rectangle argument of [12, Lemma 3.12]. Cut the enclosing rectangle into the prescribed pieces. Smooth-gradient seam errors have vanishing primary reference cost. Each partial observation has the limiting variance upper bound a times its energy. Subtracting the bounds for the other pieces from the saturated enclosing observation gives saturation on each piece. Energy density and the primary inverse-form approximation extend this conclusion to the required smears. For a component phase of nonzero integer total charge, precision monotonicity bounds its characteristic function by the enclosing-rectangle one. Theorem 2.5, also with mean-free sources, therefore gives the independent uniform fractional mean on each free component.

Lemma 3.3 (Band density bounds). *For every fixed $\lambda > 0$, $q \geq 0$, and $1 \leq p < \infty$,*

$$\limsup_{\ell \downarrow 0} \limsup_{n \rightarrow \infty} \mathcal{P}_n(\widehat{W}^p(1 + |S|)^q) < \infty, \quad S = (h, u), \quad (3.4)$$

where u is the uniform primary probability on U . In the first two limits of (3.3), fixed density smears under the band law converge with every fixed moment to the Gaussian field of precision

$$T_\lambda = -\Delta_U/a + \lambda 1_K.$$

For a smooth shift g vanishing near K , its weak Neumann Laplacian test has limiting variance $a\mathcal{E}_U(g)$, including its free-face terms. Its limiting fourth moment is bounded uniformly for $\lambda \geq \lambda_0 > 0$.

Proof. At a fixed band partition, the free limits and the uniform fractional mean make translation summation converge to integration of the constant against $dc/(2\pi)$. To justify unbounded insertions, let $\bar{h}_K = |K|^{-1} \sum_d |d| h_d$. The band energy is at least $\lambda |K| \bar{h}_K^2/2$, while $S - \bar{h}_K$ is a mean-free test of bounded primary reference cost. Lemma 2.3 supplies all its moments and fixed linear exponential moments. Summing the Gaussian translation tail gives uniform integrability for every fixed polynomial insertion. This also proves convergence of the normalizing constants and of the weighted moments at each fixed partition.

After that limit, integrate the constant first. On the mean-free Neumann field the remaining quadratic penalty is the squared $L^2(K)$ norm of its cell projection after subtracting its K -average. Its covariance C_ℓ satisfies $\sup_\ell \|C_\ell\|_{\text{HS}} < \infty$: the square's inverse Neumann eigenvalues are square summable; cutting a permitted domain into finitely many rectangles gives the same bound by the compact energy inclusions of [12, Lemma 3.12]. If t_j are the eigenvalues of λC_ℓ , the logarithm of the determinant ratio in the normalized p -th weight moment is

$$\frac{1}{2} \sum_j \{p \log(1 + t_j) - \log(1 + pt_j)\} \leq C_p \sum_j t_j^2.$$

The constant integrals are bounded positive factors. Poincare's inequality and control of the constant by the band give, uniformly under cell refinement,

$$\|v\|_{L^2(U)}^2 \leq C_\lambda \left(a^{-1} \mathcal{E}_U(v) + \lambda \sum_d |d| v_d^2 \right).$$

Thus every fixed moment of S under the continuum p -weighted law is bounded. This proves (3.4) in its stated iterated-limsup sense. It does not assert a lattice quadratic-exponential moment.

Strong convergence of the cell projections in $L^2(K)$, weak energy compactness, and smooth recovery sequences give convergence of the inverse forms to T_λ^{-1} , exactly as in [12, Lemma 3.1]. These prove the finite Gaussian limits. If g vanishes near K , then in the weak Neumann sense $T_\lambda(ag) = -\Delta_U g$; hence its test variance is $a\mathcal{E}_U(g)$. Gaussian covariance decreases with λ , which proves the fourth-moment assertion. $\square$

Metric-graph Gaussian sign clusters and their Markov exploration are developed in [10, Section 3 and Lemma 3.2]; the integer-restricted Gaussian cable construction appears in [1, Section 7, Proposition 7.1]. For Bessel heights, [14, Section 2.2, Remark 1] gives a cable construction using conditioned differences of Poisson processes. Here we use Gaussian cables on the finite series approximants and prove the uniform primary-observation estimates needed to pass to Bessel weights. On the N -subdivided lattice use the Gaussian chain weights from Lemma 2.2. Give each quadratic bond its independent Brownian bridge conditional on endpoint heights. A bond is open when its bridge does not hit zero. Conditional on magnitudes and this open graph, nonzero components have independent fair signs. Exploration reveals a bridge only until its next vertex or its first zero. The unobserved remainder starts at zero and has a shorter bridge time, hence increased precision. Conditional on a completed exploration, the unvisited law before band weighting is a centered Gaussian lattice law with additional zero pins and tightened outgoing segments. These are the finite-dimensional bridge facts used in [12, Equation (3.8)]. More explicitly, the remaining transition kernel at a stopped segment is proportional to $p_t(0, h_y)$, with t no larger than its original time, regardless of the nonzero value from which the revealed part began. That revealed value contributes only to the exposed part of an observation and, after band weighting, to its linear cross-term. For a centered unvisited test, setting visited primary coordinates to zero is its zero extension, not an assertion that their exposed values were zero.

Only primary heights contribute to smears or cluster masses. A cluster carrying primary mass is counted once, for example by its least primary vertex. A cluster of positive macroscopic diameter contains a primary path with bounded successive steps. Clusters confined to a single original edge have vanishing scaled diameter.

For clarity, the uniform estimate behind this use of the cable is elementary. With $q = b/(2N)$, for $|t| \leq R$,

$$\log \frac{\sum_{j \in \mathbb{Z}} q^{j^2} e^{2\pi t j}}{\sum_{j \in \mathbb{Z}} q^{j^2}} \leq C_{b,R} t^2 / N \quad (N \text{ sufficiently large}). \quad (3.5)$$

Group opposite j 's and use $\cosh(2\pi t j) - 1 \leq C_R t^2 j^2 e^{2\pi R|j|}$; the resulting series is $O_{b,R}(t^2/N)$. Extending a unit-edge flow through a full chain therefore costs at most a constant times its squared coefficient. Additional internal zero pins or shorter cable segments only improve this centered exponential comparison. In particular it is not the bare series resistance that is used.

Lemma 3.4 (Primary path anchoring). *After exposure of a cluster of scaled diameter at least $r > 0$, the centered posterior moments of every fixed primary density smear are bounded in terms of its unit-edge reference cost, uniformly in subdivision. The remaining full-component mean has reference cost at most V_r . If $\mathcal{D}$ consists of band cells touched by the exposed path, and $\ell < r/C$, the reference cost of $\sum_{d \in \mathcal{D}} |d| b_d h_d^U$ is at most*

$$C\ell^2 \sum_{d \in \mathcal{D}} |d| b_d^2. \quad (3.6)$$

Here averages retain the full primary cell denominator. The same conclusions hold for exploration from an entire filled starting region. A probability average in a cell reached by a path from a fixed multiple of its diameter away has a uniformly bounded posterior reference cost.

Proof. Extend variations on unvisited primary vertices by zero on the visited primary set. Apply the ordinary unit-edge path-Poincare argument of [12, Lemma 3.2]: in a fixed-factor enlargement of a touched cell, stop the primary path after it spans a comparable cell distance. One coordinate spans a positive fraction of the lattice levels, giving anchored rows or columns. One-dimensional telescoping on those rows, followed by rectangle Poincare, bounds the cell's scaled L^2 norm by $C\ell^2$ times its local unit-edge energy. The enlargements have bounded overlap. Duality with flows gives (3.6). A macroscopic anchored rectangle and global Poincare control the full-component mean.

At a free wall or a permitted junction, use the bounded union of comparable rectangles within the same component. Their face Poincare inequalities control differences of rectangle means. The no-opposite-only convention ensures these rectangles are joined through faces; one anchored rectangle anchors the union. A filled starting region provides the initial zeros. For a normalized cell average its squared L^2 density is $O(\ell^{-2})$, canceling the local ℓ^2 .

These are inequalities for the primary reference network. Extend the optimizing flows through chains and use (3.5), imposing the posterior constraints and increased precisions by Lemma 2.2. One may insert the removed internal vertices at zero, ground them, and tighten the first surviving segment; the resulting centered quadratic law is precisely the unvisited posterior. The flow identity involves only the unvisited test, since all inserted and visited coordinates in this comparison are zero. This transfers the inequalities to centered posterior moments, uniformly in N . $\square$

3.3 Polynomial damping and asymptotically fair signs

Write the sigma-finite cable measure as $\nu_n(d\omega)\mathcal{R}_\omega(d\sigma)$, where ω is signless data and $\mathcal{R}_\omega$ gives fair independent cluster signs. For a primary smear f , put

$$M_C(f) = \sum_{x \in C} |h_x| f_x, \quad m_C = M_C(u), \quad Q = \sum_C m_C^2.$$

All nonnegative calculations can first be made by Tonelli, before finiteness of the sigma-finite integrals is known. For a mean-free smear f , let $\mathcal{R}(f)$ denote its minimum unit-edge squared flow cost.

Lemma 3.5 (Bins, reinsertion, and polynomial tails). *For $t \geq 1$ and every integer $p \geq 1$,*

$$\begin{aligned} \nu_n(Q \leq 4t^2) &\leq C(1+t), \\ \int_{Q \leq 4t^2} \left(\sum_C M_C(f)^2 \right)^p d\nu_n &\leq C_p(1+t) \left(t \left| \sum_x f_x \right| + \mathcal{R}(f - u \sum_x f_x)^{1/2} \right)^{2p}. \end{aligned} \quad (3.7)$$

After exposure of a cluster of diameter at least r , the complement squared-mass sum Q_U has conditional moments of every fixed order bounded uniformly by constants depending on that order and r . Consequently, for every nonnegative intrinsic completed-cluster cost $I(C)$, and all sufficiently large $t = t(r)$,

$$\int \sum_{\text{diam } C \geq r} I(C) 1_{\{m_C \leq 2t\}} d\nu_n \leq 2 \int 1_{\{Q \leq 8t^2\}} \sum_{\text{diam } C \geq r} I(C) 1_{\{m_C \leq 2t\}} d\nu_n. \quad (3.8)$$

For every $j > 0$ there is a sufficiently large M such that

$$\mathcal{P}_n \left((1 + |S|)^{-M} 1_{\{t^2 < Q \leq 4t^2\}} \right) \leq C_j(1+t)^{-j}. \quad (3.9)$$

All estimates are uniform in large subdivision and fine primary mesh on the fixed geometry.

Proof. Given signless data, $\mathbb{E}_\sigma S^2 = Q$. On $Q \leq 4t^2$, Chebyshev gives high probability of $|S| \leq Ct$. For $Y = (h, f)$, the Rademacher fourth-moment bound and Paley–Zygmund give a fixed positive probability of

$$|Y| \geq c \left(\sum_C M_C(f)^2 \right)^{1/2}.$$

Choose C so the two events intersect with a fixed positive probability. On every translation orbit, $|S| \leq Ct$ allows at most $C'(1+t)$ translations. A mean-free Y is translation invariant and has moments of every order bounded by Lemma 2.3. This gives arbitrary polynomial bin tails, and integration gives (3.7). For nonzero total charge, use $Y = (h, f - u \sum f) + S \sum f$ on the same intersection.

After a macroscopic cluster is exposed, let S_U be the complement signed mean. Lemma 3.4 gives every conditional moment of S_U , uniformly. Jensen in its remaining fair signs yields

$$Q_U^p \leq \mathbb{E}_\sigma |S_U|^{2p}.$$

Thus $\mathbb{P}(Q_U \leq 4t^2 \mid \mathcal{F}_C) \geq 1/2$ for sufficiently large t . If $m_C \leq 2t$, this event implies $Q \leq 8t^2$. Multiply by $I(C)1_{\{m_C \leq 2t\}}$, condition, and sum by the least-primary-root disintegration. This proves (3.8), also for initially extended nonnegative integrals.

For (3.9), choose a small fixed diameter threshold r and finitely many separated probability box smears p_j so every cluster of diameter below r misses one box. If small clusters contribute at least $t^2/2$ to Q , assign each a missed box. Since $M_C(p_j - u) = -m_C$ there, one mean-free squared-mass sum is at least ct^2 . Its polynomial bin tail, of an arbitrarily high order, controls this part.

Otherwise the macroscopic clusters supply the majorant

$$1_{\{t^2 < Q \leq 4t^2\}} \leq \sum_{\text{diam } C \geq r} \frac{2m_C^2}{t^2} 1_{\{m_C \leq 2t\}} 1_{\{Q > t^2\}}.$$

Expose C , and write $S = \sigma_C m_C + S_U$. If $m_C \geq t/2$, separate $|S_U| \leq m_C/2$ from its complement. Polynomial damping handles the first event; arbitrarily high conditional moments of S_U handle the second. If $m_C < t/2$, the condition $Q > t^2$ forces $Q_U > 3t^2/4$, again giving arbitrary polynomial decay. Precisely, retain the intrinsic cost

$$I_t(C) = \frac{2m_C^2}{t^2} 1_{\{\text{diam } C \geq r, m_C \leq 2t\}}.$$

After dropping the original upper-bin restriction, the conditional expectation of $(1 + |\sigma_C m_C + S_U|)^{-M} 1_{\{m_C^2 + Q_U > t^2\}}$ is at most $C_K(1+t)^{-K}$, uniformly on this intrinsic cutoff, by the two cases just proved. This calculation is under the unweighted base disintegration. Reinsertion gives

$$\int \sum_C I_t(C) d\nu_n \leq 2 \int 1_{\{Q \leq 8t^2\}} \sum_C I_t(C) d\nu_n \leq 32 \nu_n(Q \leq 8t^2) = O(1+t).$$

Choose $K > j+1$. No weighted conditioning or count of microscopic clusters enters this estimate. $\square$

Lemma 3.6 (Weighted cluster moments). *At fixed λ , all fixed moments of squared primary cluster-mass sums and signlessly selected signed partial sums are bounded in the first two limits of (3.3). For a band cell d , with $D_\ell^2 = 1 + \log(1/\ell)$,*

$$\mathbb{E}_W \left(\sum_C M_C(d)^2 \right)^p + \mathbb{E}_W |d_*|^{2p} \leq C_{p,\lambda} D_\ell^{2p} + o_n(1). \quad (3.10)$$

Here d_* is any signed selection determined by signless data, with nonnegative coefficients bounded by the full cell smear. Fixed observation boxes have a fixed constant in place of D_ℓ .

Proof. Let F be a squared-mass root or such a signed sum and let A_t be a dyadic bin of Q . Khintchine's inequality and (3.7) give $\mathcal{P}_n(|F|^{2p}1_{A_t}) \leq P_p(t)D^{2p}$, where P_p is a polynomial and D the relevant reference cost scale. Holder with exponents 4, 4, 2 gives

$$\begin{aligned} \mathcal{P}_n(\widehat{W}|F|^p1_{A_t}) &\leq \mathcal{P}_n(\widehat{W}^4(1+|S|)^M)^{1/4} \\ &\quad \times \mathcal{P}_n((1+|S|)^{-M}1_{A_t})^{1/4} \mathcal{P}_n(|F|^{2p}1_{A_t})^{1/2}. \end{aligned}$$

The two polynomial weights cancel exactly. Use Lemma 3.3 and choose the decay order in (3.9) after p , so that the dyadic sum and its tail converge uniformly. The lowest bin is bounded directly by (3.7).

The primary reference cost of $h_d - S$ is $O(1 + \log(1/\ell))$. On a containing rectangular patch, sum the inverse Neumann cosine eigenvalues up to frequency ℓ^{-1} ; this costs $O(\log(1/\ell))$. Above that frequency Parseval and the squared density bound $O(\ell^{-2})$ give a bounded remainder. Component Poincare compares the patch and component means. This proves (3.10). An auxiliary independent sign sample has the same estimates, by exchanging the old and new samples in the base measure. $\square$

Lemma 3.7 (Macroscopic sign resampling). *At fixed $r, \lambda > 0$, replacing the signs of all clusters of diameter at least r by independent fair signs changes the band law by total variation tending to zero as $n \rightarrow \infty$ and then $\ell \downarrow 0$. Clusters of diameter below r contribute vanishing fixed moments to bounded density smears as $r \downarrow 0$ after these limits. For nonnegative fixed density smears f, g ,*

$$\limsup_{\ell \downarrow 0} \limsup_{n \rightarrow \infty} \mathbb{E}_W \sum_{\text{diam } C \geq r} M_C(f)M_C(g) \leq (f, T_\lambda^{-1}g). \quad (3.11)$$

Every fixed moment of $\sum_{\text{diam } C \geq r} M_C(f)^2$ has a limiting upper bound uniform for $\lambda \geq \lambda_0 > 0$.

Proof. On $Q \leq t^2$, expand the difference of the two quadratic band energies in the original and new signs. Orthogonality of Rademacher monomials bounds its squared base integral by

$$C\lambda^2 \int 1_{\{Q \leq t^2\}} \sum_{\text{diam } C \geq r} \sum_{C' \neq C} \left(\sum_d |d| M_C(d) M_{C'}(d) \right)^2 d\nu_n.$$

Expose C , retain the intrinsic condition $m_C \leq t$, and drop the complement bin restriction. Averaging the complement signs turns the inner sum into the conditional second moment of $\sum_d |d| M_C(d) h_d^U$. Lemma 3.4 bounds it by $C\ell^2 \sum_d |d| M_C(d)^2$. Reinsert a bin using (3.8), and apply (3.7). Thus

$$\mathcal{P}_n \left(1_{\{Q \leq t^2\}} |\log W(\sigma') - \log W(\sigma)|^2 \right) \leq C_{\lambda, t, r} \ell^2 D_\ell^2 + o_n(1).$$

The bin has finite base mass. The normalized densities have uniform L^p bounds there. Split according to whether the logarithmic difference exceeds a fixed $\delta > 0$, use Holder on that exceptional set and $|e^x - e^y| \leq (e^\delta - 1)(e^x + e^y)$ on its complement. Then let $\delta \downarrow 0$. Lemma 3.6 removes the bin cutoff, including against observables with the stated high moments. For this last step, use the enlarged base space carrying the old and new independent macroscopic sign samples, with the small-cluster signs shared. Swapping the two samples preserves the base measure and the signless variable Q . Thus the old and new normalized densities have identical Q -tails, and for every fixed p ,

$$\int_{\{Q > T\}} |\widehat{W}(\sigma) - \widehat{W}(\sigma')| \leq 2T^{-p} \mathbb{E}_W Q^p.$$

The right side is uniformly small in the first two limits. For auxiliary-sign observables the unweighted bin moment estimate is unchanged. For each density use its own component mean in both damping

factors; swapping the samples gives the bound for the other density. Hence the weighted Holder estimate controls those observables under either density.

Partition U into r -cells. A cluster of diameter below r meets only boundedly many neighboring cells. For a bounded density, Cauchy–Schwarz and (3.7) bound its signed second moment on a fixed bin by $C_t r^2(1 + \log(1/r))$. Higher fixed moments are bounded there. Interpolation, followed by the weighted tail transfer, gives their vanishing.

Resample first at a threshold below r , remove the smaller clusters, and use Lemma 3.3. Fair-sign second moments are the cluster cross sums; nonnegative smears make their summands nonnegative. This proves (3.11). Jensen in the fair signs similarly bounds the p -th moment of a squared-mass sum by the Gaussian $2p$ -th smear moment. The latter is uniformly bounded for $\lambda \geq \lambda_0$, since the Gaussian covariance decreases with λ . $\square$

3.4 Tilt errors and vanishing reach

For an exploration under the band law, write $h_d = d_* + h_d^U$. Relative to the centered posterior with the restricted band form, its remaining linear tilt is

$$L = -\lambda \sum_d |d| d_* h_d^U.$$

Lemma 3.4 bounds its reference cost by $\epsilon^2 = C_0 \lambda^2 \ell^2 \sum_d |d| d_*^2$. For a single macroscopic cluster C , define $\epsilon_C^2 = C_0 \lambda^2 \ell^2 \sum_d |d| M_C(d)^2$ and $E_2 = \sum_{\text{diam } C \geq r} \epsilon_C^2$. Weighted Jensen and (3.10) give, for every fixed p, λ, r ,

$$\mathbb{E}_W \epsilon^{2p} + \mathbb{E}_W E_2^p \leq C_{p,\lambda,r} \ell^{2p} D_\ell^{2p} + o_n(1). \quad (3.12)$$

If $\epsilon \leq 1$, the centered posterior μ satisfies

$$1 \leq \mu(e^L), \quad \mu(e^{2L}) \leq e^{C\epsilon^2}, \quad \mu((e^L - 1)^2) \leq C'\epsilon^2$$

by (3.5) and precision comparison. Cauchy–Schwarz with fixed-cost fourth moments shows that tilting changes means and second moments of fixed unexposed smears by $O(\epsilon)$. For an inversion-symmetric event E , symmetry also gives the relative estimate

$$\mu_L(E) \geq e^{-C\epsilon^2} \mu(E). \quad (3.13)$$

Indeed $\mu(e^L | E) = \mu(\cosh L | E) \geq 1$. This estimate does not depend on the size of $\mu(E)$.

Large tilts are truncated rather than estimated by a Gaussian tail. Here is the exact summation needed for the size-biased exploration. For comparison boxes B, i , put

$$Q_B = \sum_{\text{diam } C \geq r} M_C(B)^2, \quad Q_i = \sum_{\text{diam } C \geq r} M_C(i)^2.$$

Since the number of clusters with $\epsilon_C > 1$ is at most E_2 , pointwise

$$\sum_{\text{diam } C \geq r} M_C(B) 1_{\{\epsilon_C > 1\}} (1 + h_i^2 + M_C(i)^2) \leq \sqrt{Q_B E_2} (1 + h_i^2 + Q_i). \quad (3.14)$$

Its expectation vanishes by Holder, the actual band moments, and (3.12). This controls the actual unvisited contribution because $|h_i^U| \leq |h_i| + M_C(i)$ after one-cluster exposure. Centered comparison terms have the fixed-cost moment bounds. For good tilts, Cauchy–Schwarz gives

$$\sum_C M_C(B) \epsilon_C (1 + \epsilon_C) (1 + M_C(i)) \leq \sqrt{Q_B} (1 + \sqrt{Q_i}) (\sqrt{E_2} + E_2),$$

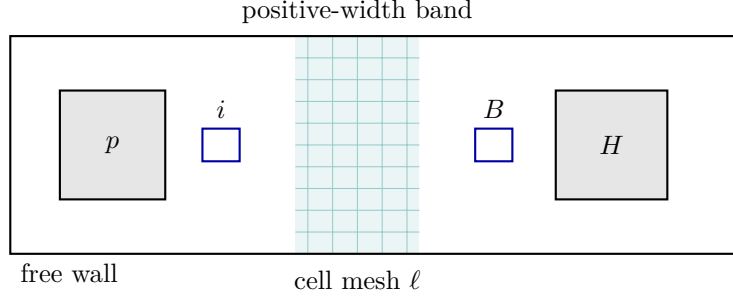


Figure 1: The band separates two positive-area pin regions. The test boxes lie away from the band. Its cell mesh ℓ is refined while the physical band width stays fixed; the band strength is increased only afterwards. Hard pins constrain all primary height vertices in their regions.

whose expectation also vanishes. For exterior exploration, actual band moments and the vanishing moments of ϵ control the discarded event $\epsilon > 1$ directly. These estimates use no quadratic-exponential moment of a Bessel height and no subdivision-dependent bare variance.

We also need the size-bias inequality

$$\mathbb{E}[X^2|Y|] \geq \mathbb{E}X^2 \mathbb{E}|Y| \quad (3.15)$$

for real linear tests in a proper centered law with band precision. On each subdivided Gaussian lattice it follows from the minimal characteristic logarithmic Hessian: for $\phi(t) = \mathbb{E} \cos(tY) > 0$,

$$\mathbb{E}[X^2 \cos(tY)] \leq \mathbb{E}X^2 \phi(t).$$

Subtract the identity at zero and integrate against $2 dt/(\pi t^2)$, using $|y| = (2/\pi) \int_0^\infty (1 - \cos ty)t^{-2} dt$. Tonelli proves the inequality, and the fixed-graph moment limit transfers it to primary heights. This is the size-bias part of [12, Lemma 3.6].

Proposition 3.8 (Vanishing reach). *Under the band law, the probability that an exploration starting in a fixed positive-area region on one side of a separating positive-width band reaches a fixed closed region on the other side tends to zero in (3.3). The assertion includes exploration from a filled exterior or from a primary hard-pin region before imposing its pin.*

Proof. Choose finite small-cell coverings i on the starting side and B on the target side, away from the band and with positive clearances. Enlarge overlapping cells slightly so every crossing path meets both coverings after rounding; see Figure 1. Concretely, first choose two separator strips, clipped to the free component, with a fixed clearance $\eta > 0$ from the band and the filled starting and target regions. Then cover them by cells of radius $\delta \ll \eta$. A reaching path contains, in a fixed-factor enlargement of the touched cell, a segment spanning $c\delta$. The path-Poincare constant on this rescaled patch depends only on the fixed finite face and corner geometry, not on δ ; at a wall use the adjacent face-connected rectangular pieces of that same component. The primary path bound gives a uniform upper bound C_0 for the centered remaining average in a reached cell. A logarithmic trial supported in a fixed ball away from the band gives continuum prior variance at least $ca \log(1/\delta) - C$ for a cell of radius δ . Cut the trial off at a radius $\rho < \eta/4$ fixed before δ . It pays no band mass. At a free wall or corner restrict it to the component; this is an admissible Neumann energy function. Its cell average is bounded below by $c \log(\rho/\delta) - C$, and its energy is at most $C \log(\rho/\delta) + C$, uniformly over the bounded cell shapes. Thus shrinking the cells increases the prior lower bound without increasing the path-anchoring constant. Choose the covering cells so

small that the prior variance exceeds $C_0 + 2c_0$, for some $c_0 > 0$, uniformly in λ . At each fixed λ , band convergence gives the strict gap c_0 on fine meshes.

For exterior exploration write $h_B = e_B + u_B$, exposed plus unexposed. The centered posterior is obtained by zero-pinning visited vertices and increasing cable precision. Thus its variance for u_B is at most the full prior variance v_B , and at most $v_B - c_0$ on a visit to B . Restoring the tilt changes the averaged second moment and cross term by $o(1)$, by the good- and bad-tilt estimates above. The identity $\mathbb{E}_W h_B^2 = v_B$ therefore gives $c_0 \mathbb{P}_W(B \text{ reached}) \leq \mathbb{E}_W e_B^2 + o(1)$. All contributing clusters are macroscopic, with a threshold fixed by the separation from the starting set. Sign resampling, with that same signless selection, yields

$$c_0 \mathbb{P}_W(B \text{ reached}) \leq \mathbb{E}_W \sum_{C \text{ meeting the starting set}} M_C(B)^2 + o(1). \quad (3.16)$$

For the second deficit, bias the prior by $|h_x|$, where $x \in B$, and expose only its cluster. The pre-exposure second moment at i is at least v_i by (3.15). The bias is measurable at completed exploration, so cancels from the conditional complement law. Its centered remaining variance is at most v_i , and at most $v_i - c_0$ when the cluster visits i . Multiply the total second-moment comparison by $\mathbb{E}|h_x|$ and the primary averaging weight of x , then sum over $x \in B$. This produces precisely the weight $M_C(B)$, not a count of microscopic clusters. Only clusters reaching the band or i cause a nonzero tilt or exposed term; the separating gaps make all of them macroscopic. The estimates preceding the proposition give

$$c_0 \mathbb{E}_W \sum_{C: C \cap i \neq \emptyset} M_C(B) \leq \mathbb{E}_W \sum_C M_C(B) M_C(i)^2 + o(1). \quad (3.17)$$

Points with $\mathbb{E}|h_x| = 0$ contribute zero.

As $\lambda \rightarrow \infty$, the inverse forms T_λ^{-1} decrease to the inverse with every energy function zero on K . Weak energy compactness and recovery by such functions prove this statement directly. The hard band separates the sides, so its cross covariance is zero. Equation (3.11) therefore makes $\mathbb{E} \sum_C M_C(B) M_C(i)$ vanish in the prescribed order. Truncate first at $\max_C M_C(i) \leq M$; on this event the right side of (3.17) is at most M times that cross sum. Its remaining tail vanishes uniformly as $M \rightarrow \infty$, since

$$\sum_C M_C(B) M_C(i)^2 \leq \sqrt{Q_B} Q_i$$

and Lemma 3.7 supplies arbitrarily high moments uniform toward large λ . Consequently the first-mass sum on the left of (3.17) vanishes. Every starting-set cluster meeting B visits one of the finitely many cells i . Truncate $\max_C M_C(B)$ in (3.16) and use these first-mass bounds; its tail is controlled by the moments of Q_B . This proves vanishing reach.

All exploration calculations were on finite subdivided Gaussian graphs. Their primary bounds are uniform in large N , or are passed first at a fixed primary graph. Thus no continuum-time cable realization of the limiting Bessel law is needed. $\square$

3.5 Mixed limits and relative hard-pin comparison

Proof of Theorem 3.1. On a component meeting p , start with a smooth shift g vanishing near the entire pin but allowed to meet the free walls. Choose a separating band between p and its support, with fixed gaps. The band may be a union of strips clipped to the component. Use the ordinary mixed-domain geometry from [12, Proposition 3.13]; Lemma 3.4 already verifies its reference inequalities at faces and junctions.

For the weak Neumann Laplacian test of g , the band variance is $a\mathcal{E}_U(g)$. Explore from every primary vertex of p . On success, the exposed test is zero and the centered posterior has at least the desired primary pin, possibly further zero pins, tighter chains, and restricted band precision. Its variance is therefore at most that under the desired massless hard- p law. The tilt errors vanish as above. On failure the actual second moment is bounded by its fourth-moment root times the square root of the failure probability. Lemma 3.3 bounds that fourth moment uniformly after the first two limits and toward large λ . Proposition 3.8 therefore gives

$$\liminf_{n \rightarrow \infty} \text{Var}_p(h, Ag_n) \geq a\mathcal{E}_U(g).$$

Run the argument on an arbitrary primary subsequence. The free variance upper bound gives the opposite limsup. The free logarithmic transform upper bound and the centered-variance lower bound under real tilt then give the matching Gaussian Laplace limit.

Weak free-face terms may first be approximated by bounded density smears in primary reference cost, preserving their gap from p and the band. The real mixed inverse-form approximation and density of smooth shifts vanishing near p , from [12, Lemma 3.12], extend the result to all asserted tests. Lemma 2.3 controls the errors independently of subdivision. It supplies uniform integrability for every fixed linear exponential and polynomial insertion.

Unpinned components were handled before Lemma 3.3. Finally integrate a fixed finite-rank quadratic penalty against these massless limits. On pinned components the preceding moment bounds suffice. On unpinned components sum translations and use the independent uniform fractional mean, with the confining penalty controlling the constants exactly as in the band-density proof. This proves the theorem, including its limit order and its statements for rounded geometry sequences. $\square$

We now prove convergence of the interaction in (3.1); this is where a relative, rather than absolute, estimate is essential. Write U for the original finite box-average penalty and W for a separating band penalty. Let $Z_{M,Q}$ be the partition sum with penalty M and primary hard pin Q . Cancellation gives

$$D_U - D_W = \log \frac{Z_{U,H \cup p}}{Z_{W,H \cup p}} + \log \frac{Z_{U,\emptyset}}{Z_{W,\emptyset}} - \log \frac{Z_{U,H}}{Z_{W,H}} - \log \frac{Z_{U,p}}{Z_{W,p}}. \quad (3.18)$$

Each ratio changes only a fixed finite-rank confining penalty on one fixed massless pinned or unpinned domain. Its common normalization cancels, so Theorem 3.1 gives its Gaussian limit at each fixed band partition and strength. The real Gaussian determinant theorem [12, Proposition 3.14] identifies the corresponding interaction difference.

Under the unpinned band law, explore all clusters meeting p . On the event $\mathcal{S}$ that H is untouched, the centered posterior is obtained from the band law conditioned on $h_p = 0$ by further primary or internal zero pins and increased precision. Its probability of $h_H = 0$ is at least $\mu_W(h_H = 0 \mid h_p = 0)$. For $\epsilon \leq \delta \leq 1$, the actual tilted posterior therefore satisfies

$$\mu_W(h_H = 0 \mid \mathcal{F}) \geq e^{-C\delta^2} \mu_W(h_H = 0 \mid h_p = 0).$$

Averaging gives the relative estimate

$$e^{-D_W} \geq e^{-C\delta^2} \mathbb{P}_W(\mathcal{S} \cap \{\epsilon \leq \delta\}). \quad (3.19)$$

It remains valid however small the two pin probabilities are. By precision monotonicity $D_W \geq 0$. First the tilt vanishes at fixed λ , then reach vanishes as λ increases. Sending $\delta \downarrow 0$ proves $D_W \rightarrow 0$ in (3.3).

The same vanishing holds for the real Gaussian interaction D_W^g . One may use the real-Gaussian proof of [12, Equation (3.47)]: replace pin probabilities by joint densities at zero, use the same centered precision comparison, and average small-box pin probabilities before dividing by their volumes and applying Fatou. The ordinary Gaussian localization theorem passes the separated-pin determinants to the continuum. Thus both band interactions vanish. Equation (3.18) proves

$$D_n(H, p) \longrightarrow D^g(H, p). \quad (3.20)$$

No individual microscopic pin probability has been assigned a positive continuum limit.

3.6 Regression at typical pin data and restoration

Proof of the remaining assertions of Theorem 3.2. Let X_n be a fixed real smear, $m_n(b^p) = \mathbb{E}(X_n \mid h_p = b^p)$, and $v_{p,n} = \text{Var}(X_n \mid h_p = 0)$. At prescribed primary data, the subdivided law is an affine Gaussian lattice law with the same quadratic part as at zero data. Lemma 2.2, including an additional real tilt, and then its fixed-graph limit give

$$\text{Var}(X_n \mid h_p = b^p; \text{tilt } tX_n) \geq v_{p,n}. \quad (3.21)$$

Choose the first J smooth density probes on p , and let $L_n^{(J)}$ be their linear combination with the limiting Gaussian regression coefficients. Remove redundant directions or invert on their Gram-matrix range. Conditional variance decomposition gives the exact bound

$$\begin{aligned} \mathbb{E}|m_n - L_n^{(J)}|^2 &= \mathbb{E}|X_n - L_n^{(J)}|^2 - \mathbb{E} \text{Var}(X_n \mid h_p) \\ &\leq \mathbb{E}|X_n - L_n^{(J)}|^2 - v_{p,n}. \end{aligned} \quad (3.22)$$

Theorem 3.1 identifies both limiting variances. As $J \rightarrow \infty$, totality of the pin probes makes the right side tend to zero: in the Gaussian energy space it is the squared difference between the full pin projection and the J -probe projection of the representer of X . Thus the conditional mean is approximated in L^2 by the finite Gaussian linear predictions.

Integrating (3.21) twice in t gives

$$Y_n := \mathbb{E}(e^{X_n} \mid h_p) \geq \exp\{m_n + v_{p,n}/2\}.$$

Both sides have limiting expectation $e^{v/2}$, where v is the full limiting variance. For the right side, use (3.22) and uniform exponential integrability: $\mathbb{E}e^{qm_n} \leq \mathbb{E}e^{qX_n}$ by Jensen for every fixed real q . Their nonnegative difference tends to zero in L^1 , proving the conditional Laplace assertion. Apply the same argument to a countable dense set of linear combinations of any fixed test list. Subsequence extraction and conditional exponential moment bounds give tightness and identify all conditional finite-dimensional weak limits. This also proves conditional convergence of fixed smooth quadratic penalties.

For the pin ratio, precision comparison at prescribed b^p shows that every normalized conditional positive-penalty ratio is at most one and decreases when the penalty is increased. Letting a full coordinate pin on H tend to infinity yields $0 < r_{H,n} \leq 1$. Its definition gives exactly $\mathbb{E}r_{H,n} = e^{-D_n}$.

Let Q_J be increasing finite smooth quadratic penalties exhausting the pin on H , and put

$$R_{J,n}(b^p) = \frac{\mathbb{E}(Q_J \mid h_p = b^p)}{\mathbb{E}(Q_J \mid h_p = 0)}.$$

Then $r_{H,n} \leq R_{J,n} \leq 1$. For fixed J , the conditional weak convergence and the centered hard- p limit identify $R_{J,n}$ with its Gaussian counterpart. Moreover,

$$\mathbb{E}(R_{J,n} - r_{H,n}) = \mathbb{E}R_{J,n} - e^{-D_n}. \quad (3.23)$$

The interaction limit (3.20) and the finite-probe determinant approximation in [12, Proposition 3.14] make the right side tend to zero as $J \rightarrow \infty$ after $n \rightarrow \infty$.

More explicitly, let s_j be the singular values of the cross projection between the Gaussian pin spaces of p and H . Positive separation gives $\sup_j s_j < 1$ and $\sum_j s_j^2 < \infty$. In canonical standard Gaussian pin coordinates ξ_j ,

$$r_H^g = \exp \left\{ -\frac{1}{2} \sum_j \frac{s_j^2}{1 - s_j^2} \xi_j^2 \right\} > 0 \quad \text{almost surely,} \quad \mathbb{E} r_H^g = \prod_j \sqrt{1 - s_j^2} = e^{-D^g}.$$

Finite probes exhaust these convergent expressions. Markov's inequality in (3.23) therefore proves the asserted finite-observation approximation of the full ratio. In particular

$$\lim_{\eta \downarrow 0} \limsup_{n \rightarrow \infty} \mu_n(r_{H,n} \leq \eta) = 0. \quad (3.24)$$

Finally a fixed finite linear tilt has uniformly bounded L^q density for every fixed finite q : its denominator is at least one, and the mixed Laplace bounds control its numerator's moments. Holder transfers the preceding convergence statements to the tilted law. Here $r_{H,n}$, D_n , and the conditional functions being sampled keep their centered definitions; in particular we do not claim $\mathbb{E}_{\mu_{n,L}} r_{H,n} = e^{-D_n}$. If a conditional transform under the tilted baseline is needed, use

$$\mathbb{E}_{\mu_{n,L}}(e^X \mid h_p) = \frac{\mathbb{E}_{\mu_n}(e^{X+L} \mid h_p)}{\mathbb{E}_{\mu_n}(e^L \mid h_p)}.$$

The denominator has a strictly positive Gaussian limit, so the established joint conditional convergence applies. This completes the theorem. $\square$

Lemma 3.9 (Restoration and two-way transfer). *Let μ_n be a cut centered high law and let ν_n restore bonds or positive forms supported inside a core H , separated from the guard p . Include both primary endpoints of every restored bond, and the support of every restored form, in the hard core pin. If Q_Δ is the positive restoration factor and*

$$r_\Delta(b^p) = \frac{\mathbb{E}_{\mu_n}(Q_\Delta \mid h_p = b^p)}{\mathbb{E}_{\mu_n}(Q_\Delta \mid h_p = 0)},$$

then

$$r_{H,n} \leq r_\Delta \leq 1, \quad \frac{d\nu_{n,p}}{d\mu_{n,p}} = \frac{r_\Delta}{\mathbb{E}_{\mu_n} r_\Delta}, \quad r_{H,n} \leq \frac{d\nu_{n,p}}{d\mu_{n,p}} \leq e^{D_n(H,p)}. \quad (3.25)$$

Events concerning random guard data whose probabilities tend to zero transfer in both directions between the two topologies. This typical-data transfer also holds when either law is sampled with a fixed finite linear tilt. The displayed identities and inequalities remain statements about the centered laws and their centered-defined ratios.

Proof. For each restored edge $e = (x, y)$, enlarge the cut law by an independent normalized chain of increments $\xi_{e,1}, \dots, \xi_{e,N} \in 2\pi\mathbb{Z}$ in height units, and put $J_e = \sum_k \xi_{e,k}$. Restoring the edge imposes $J_e = h_y - h_x$. Conditional on primary heights, its factor is the chain-convolution probability $p_{b,N}((h_y - h_x)/(2\pi))$, tending to the Bessel weight. Thus the primary factor need not itself be quadratic. On the expanded lattice, matching is the limit of the positive quadratic penalties $\exp[-t(J_e - h_y + h_x)^2/2]$.

Write Z_{b^p} and Z_0 for the expanded cut sums at the two guard data, and use a superscript Δ for restored matching and any further restored positive forms. The guard fibers are affine and centered copies of the same expanded lattice, and

$$r_\Delta(b^p) = \frac{Z_{b^p}^\Delta/Z_{b^p}}{Z_0^\Delta/Z_0}.$$

Precision comparison gives $r_\Delta \leq 1$; adding the full primary H -pin can only decrease this ratio. With both endpoints pinned, every new chain matches $J_e = 0$. Its integrated factor is the scalar $p_{b,N}(0) > 0$, independent of guard data. Hence $Z_{b^p}^{\Delta+H} = \kappa_N Z_{b^p}^H$ and $Z_0^{\Delta+H} = \kappa_N Z_0^H$, with the same κ_N . The resulting normalized ratio is exactly $r_{H,n}$, proving its lower bound. All independent-chain normalizers were identical in these sums, and both they and κ_N cancel. Internal chain variables therefore need not be pinned. Take matching penalties to infinity and then the fixed-primary-graph subdivision limit.

Bayes' formula gives the middle identity in (3.25). Its denominator lies in $[e^{-D_n}, 1]$, proving the last two bounds. The interaction has a finite limit, so the upper density bound transfers $\mu_{n,p}(A_n) \rightarrow 0$ to $\nu_{n,p}(A_n) \rightarrow 0$. Conversely, for $\eta > 0$,

$$\mu_{n,p}(A_n) \leq \mu_{n,p}(r_{H,n} \leq \eta) + \eta^{-1} \nu_{n,p}(A_n).$$

Use (3.24), sending $\eta \downarrow 0$ after the lattice limit. Fixed linear tilts transfer by Holder and the bounds for both positive and negative fixed exponential moments. All these statements concern the finite-observation sense at typical data specified above. $\square$

4 Magnetic sectors and the spin lower bound

Throughout this section, fix $b > 0$ with $a = a(b) > 0$. We prove the converse to Proposition 2.6.

Proposition 4.1 (Magnetic lower bound). *For the prescribed free-box correlation,*

$$\liminf_{n \rightarrow \infty} \frac{\log C_b(n)}{\log n} \geq -\frac{2\pi}{a}.$$

In particular, $m(b) = 0$.

Convergence of ordinary height observations does not by itself compare partition functions in different winding sectors. We make that comparison by adding confining observations on each of finitely many scales. Local Gaussian limits then control guarded bond restorations, and an exact finite expansion bounds the accumulated error. The high-law inputs are Theorems 3.1 and 3.2 and Lemma 3.9, already proved for the Skellam weights. The real Gaussian localization estimates used below are the unchanged ones in [12, Proposition 3.14].

4.1 A localized magnetic sector

Use coordinates divided by n , so the sources are $0, e_1$. Take the outer box sufficiently large that its boundary is well outside $[-8, 8]^2$ in these coordinates. There is a real multivalued function v , with periods $2\pi, -2\pi$ at the two sources, which equals a signed polar angle plus a smooth function near each source and has a single-valued lift equal to zero outside $[-2, 2]^2$. Indeed, the difference of the two angles has zero period on an outer annulus; multiply its lift there by a cutoff. Choose bounded-value branches, with bounded integer jumps. Whenever a local chart is changed, apply the same integer gauge to the lattice height and to v .

Fix a large microscopic length m_* . Choose q so that

$$m_* \leq 2^{-q}n \leq 2m_*,$$

and remove the dual height vertices in two open squares of half-side 2^{-q} about the sources, rounding their faces to lattice divisions. The new inner walls are free. The prescribed connection still has the two indicated periods in this punctured graph. Below, all observations are outside the holes and are contained in charts missing the sources.

Let B be a collection of box averages, to be specified shortly. Add the observation factor

$$\exp \left\{ -\frac{1}{2} \|Bh - Bv\|^2 \right\} \quad (4.1)$$

in the shifted sector and use target zero in the centered sector. Equivalently, average the single-valued deviation in each chart. The two targets and their connection can be chosen zero on a common outer terminal region. Lemma 2.2 shows that adding (4.1) decreases the shifted-to-centered ratio: in enlarged affine coordinates the new coordinates are $Bh - Bv$, versus Bh on the parallel centered lattice, and we are adding a positive quadratic form. No integer-valued assumption on v is used.

Here is the geometry, including the uniformity needed later. First choose a sufficiently large dyadic integer M , and then a large dyadic integer R . There is a face-connected tessellation by regular tiles and one unprocessed exterior object with the following properties.

- (i) Away from outer termination, tile size is comparable to distance to the nearer source divided by M , with distance capped at a fixed constant away from the sources. Each tile contains mass cells whose side lengths are comparable to its size divided by R .
- (ii) Patches of every fixed tile-radius used below have a bounded number of tiles, local charts missing the sources, and comparable mass-cell sizes. Their pre-pin components, including after local bond deletions, have the mixed geometry of Theorem 3.1. Guards and cores are unions of tiles; their intervening collars have width comparable to R in mass-cell units. Mass cells do not cross a guard or a tile cut.
- (iii) There are at most $C_M(1 + \log n)$ processed tiles and uniformly bounded face-graph degrees. On each fixed-radius patch, a target lift has bounded magnitude and bounded sampled gradient energy, uniformly in R, n . Its vector of mass-cell averages has norm at most a fixed polynomial in R .
- (iv) The exterior object starts beyond the support of the target, contains the remaining graph up to the outer height pin, and is never processed. An ordinary continuation of the local tessellation and of the mass cells is retained through a wide collar of its inner boundary. Local patches use this continuation and stay far from the outer pin. For dependency counting only, tiles attached to the exterior may be joined through one terminal node of degree bounded in terms of M ; local distances never use this shortcut.

To construct the tessellation, divide the sup-norm neighborhood of radius $1/4$ of each source into square rings with successive radii in ratio two, down to its hole. Subdivide a ring into squares of side its inner radius divided by M ; use a matching uniform grid in the remaining bounded region. At subdivision junctions insert small trimmed square tiles, with side a fixed fraction of the shortest incident side. Clip at the holes when necessary. Thus point contacts do not join components, and incident sectors which must be joined have a face, rather than an opposite-only junction. This is the geometry of [12, Definition 3.11]. Subdivide all tiles and trimmed parts into rectangular mass

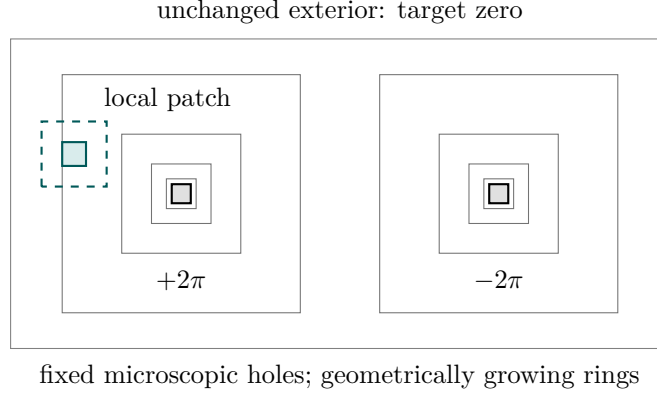


Figure 2: Schematic geometry for the winding-sector comparison. Each square ring is partitioned into tiles much smaller than its distance to the source; only one tile and its enlarged patch are shown. The construction trims tile junctions and uses collars with full interfaces. The number of processed tiles is $O_M(1 + \log n)$, while the unprocessed exterior is kept intact.

cells of the stated size, with compatible face subdivisions. Rounded coordinates retain the same adjacencies when m_* is large enough.

Choose the fixed patch, guard, skip, and enlargement radii first, and take M larger than all of them. After rescaling by the local ring radius, the patches and their possible local deletions range over finitely many shapes, up to bounded lattice rounding. Here is why this covers every integer n , not just dyadic separations. In primary lattice units the logical ring radii are $r_j = n2^{-j}$, stopped with $m_* \leq r_q < 2m_*$. Round each shared logical face once to a dual-lattice division, and reuse it on both sides. Its displacement is $O(1)$. A fixed-radius patch spans only boundedly many neighboring ring scales, so its ideal normalized shape and deletion pattern belong to finite lists. Its mesh and face-rounding errors in mass-cell units are at most

$$\frac{C_MR}{r_j} \leq \frac{C_MR}{m_*}.$$

For large m_* , all widths, incidences, and collars are preserved.

The targets also range over compact smooth families. Near a source z_0 , after scaling by $\rho_j = r_j/n$, a local lift has the form $\pm \arg z + g(z_0 + \rho_j z)$, with g smooth and $0 < \rho_j \leq 1/4$. Adjoining $\rho = 0$ closes this family in the required smooth norms; finitely many angle charts suffice. The outer patches have fixed smooth families, and bounded integer chart changes add finitely many choices. Sampled cell averages converge uniformly on these compact sets. If a common refinement threshold failed at fixed M, R , a sequence of failing patches with mesh tending to zero would have a subsequence with one ideal shape and deletion pattern and a convergent target. The admissibly rounded high-law and real Gaussian limits on that subsequence give a contradiction. Thus one fixed $m_*(R)$ makes every local estimate below uniform over all rings, all sufficiently large integer n , and all sufficiently distant outer boundaries. Guards can be enlarged by whole tile layers, also at an inner free wall.

Let Z^s, Z^0 denote the punctured, observation-weighted partition functions in the shifted and centered sectors. In the same finite graph let Z_g^s, Z_g^0 be the real Gaussian integrals with gradient precision A/a , the same observations, outer zero wall, and periods. A free component has a confining observation, so its constant is integrated rather than fixed. In comparisons of the two models, a Gaussian pin probability means the corresponding density at the prescribed coordinates, or equivalently the common shrinking-neighborhood ratio. Pin interactions and their determinant

formulas are therefore meaningful although individual Gaussian pin events have probability zero.

Proposition 4.2 (Many-scale sector comparison). *For the preceding geometry there are numbers $\eta_R \rightarrow 0$ such that, for each sufficiently large fixed R , one can choose a fixed m_* for which*

$$\left| \log \frac{Z^s}{Z^0} - \log \frac{Z_g^s}{Z_g^0} \right| \leq C_M(1 + \log n)\eta_R. \quad (4.2)$$

For these fixed choices, heights on the two inner rims, in the bounded cut coordinates just specified, are uniformly tight in n and in the outer volume. This is absolute tightness, without subtracting an additive constant.

4.2 An exact isolation expansion

Fix an order of the processed tiles. An ordinary step deletes all still-present bonds from its tile to other tiles, including the exterior object. Around its core choose a free patch of fixed tile radius, containing a separating guard and collars wider than any later support enlargement. In the current local topology, let U_o, U_c be the patch before and after this proposed deletion. Both admit pure-gauge lifts. Replace the bond factor by the scalar

$$D_i^s = \frac{Z_{U_o}^0}{Z_{U_c}^0} \frac{Z_{g,U_o}^s / Z_{g,U_o}^0}{Z_{g,U_c}^s / Z_{g,U_c}^0} \quad (4.3)$$

times the cut factor, plus their difference. In the centered calculation omit the second fraction, obtaining D_i^0 . For the real Gaussian expansion use its own centered constants in the same formula.

Selecting the difference marks this step. After a mark, skip every later step within a fixed number of tile layers, larger than all patches and guard-testing boxes. Thus marks are separated, their guard interiors are left untouched, and a subsequently considered proxy cannot depend on which of the two positive terms is selected inside a previous difference. This is a finite, exact expansion. All deviations from the unmarked sequence are supported in bounded halos of marks. At its end every surviving cross-tile bond, other than a bond wholly within the exterior, lies in such a halo or joins it to the exterior.

Normalize each sector by its no-mark term: the product of its successive proxies and the final isolated tile partition functions, including the unchanged exterior. For a specified set I of j marks, write $w(I)$ for its contribution divided by this normalization.

Lemma 4.3 (Pattern gain). *For either model and either sector, after increasing m_* at fixed sufficiently large R ,*

$$|w(I)| \leq e^{-cRj}. \quad (4.4)$$

Here and below $c > 0$ may decrease, and all fixed constants may depend on b, a, M , but not on n or the outer volume.

Proof. We separate the centered normalization, the random guards, and the local gain.

Conditional differences. Prescribe all primary heights u_i on the guards of the marks. Let K_o, K_c be the two positive partition kernels inside one guard, in the corresponding marked step. Set

$$\delta_i(u_i) = \frac{|K_o - D_i^s K_c|}{K_o + D_i^s K_c} \leq 1,$$

and use D_i^0 in the centered sector. Guard and exterior factors cancel from this expression. Conditional integration of the product of differences, followed by the triangle inequality, gives a sum of at most 2^j positive topologies. In each one the integrand has the additional factor $\prod_i \delta_i$. We have not assumed independence of the random guard data.

Centered normalization. For a positive topology G , including its ordinary proxy factors, its contribution divided by the no-mark term is at most

$$e^{Cj} \frac{Z_G^s}{Z_G^0} \mathbb{E}_{G,s} \prod_i \delta_i. \quad (4.5)$$

Only affected components are included in this display; untouched final tile factors have been canceled.

Here is the accounting behind (4.5). A centered restoration supported in a core lies between its free-patch value and its value with a separating guard pinned to zero, whether the actual exterior is absent or present. Lemma 2.2 gives the direction. For a chain representation, retain the independent chain increments and restore endpoint matching; all scalar chain normalizers are the same in these comparisons. A full primary pin on a region H containing the restored endpoints makes the restoration a fixed scalar. Therefore its free-to-guarded multiplicative gap is at most

$$\frac{\mathbb{P}(h_p = 0 \mid h_H = 0)}{\mathbb{P}(h_p = 0)} = e^{D(H,p)}.$$

Theorems 3.2 and [12, Proposition 3.14] bound this uniformly, and in fact make its logarithm exponentially small in R after sufficient microscopic refinement. Indeed, gradient energy plus cell-average mass is uniformly coercive in mass-cell units, the clearance is of order R , and local cell volumes are polynomial in R . The same cell-cutoff and Hilbert–Schmidt estimates apply to the bounded rectangular shapes here, including at free walls.

Only $O_M(j)$ such normalization comparisons are charged. Take a union containing sufficiently enlarged halos of all marks, with a further collar beyond every skip and dependence radius. Telescope separately the actual centered partition functions along the positive branch and along the no-mark branch in this union. Include the whole exterior object and its neighboring collar if surviving bonds or changes can meet it. Count every event in the union, including one clipped at its artificial boundary. At a clipped event use the proxy clipped in the same manner on both paths. Before every boundary event, the two histories agree on its entire proxy patch: the collar contains every affected or skipped step and exceeds the proxy’s dependence radius. Thus the scalar change from an original proxy to its clipped version is identical on both paths and cancels. There are $O_M(j)$ events; the exterior itself has none. Final integrals and proxies outside the affected region also cancel, while the initial centered integrals on the two paths are identical. Each remaining true-to-proxy ratio is bounded by the preceding centered sandwich. Enlarging by whole tile layers ensures that clipped comparisons still have separating guards with unchanged clearance. Artificial point contacts add no bonds. More explicitly, call this induced union U , with free artificial walls, and let α be either deletion history. For an actual deletion e , put

$$T_e^\alpha = \frac{Z_{U,e}^0 \text{ before}}{Z_{U,e}^0 \text{ after}}, \quad \tilde{D}_e^\alpha = \text{its centered proxy clipped to } U.$$

A skipped step or an open marked choice is not a deletion; a cut marked choice is a deletion and carries its proxy. The exact telescoping identity is

$$Z_{U,\text{final}\alpha}^0 \prod_{e \in \alpha} \tilde{D}_e^\alpha = Z_{U,\text{initial}}^0 \prod_{e \in \alpha} \frac{\tilde{D}_e^\alpha}{T_e^\alpha}. \quad (4.6)$$

The initial integral is common to both histories. In their quotient, changing an original proxy to a clipped one cancels at the artificial boundary, where the histories agree through a collar wider than the proxy radius. Final components outside U are identical and have no surviving bond to U ; including an attached exterior object ensures this also for terminal connections. Each product on the right has $O_M(j)$ factors, each bounded with its inverse by the centered sandwich, also up to a new free wall. An exterior object has no deletion events, and its whole inner collar has only $O_M(1)$ tiles. Thus its arbitrarily large microscopic volume adds no charge. This proves e^{Cj} without an error proportional to untouched tiles, as in [12, Lemma 9.3].

The shifted proxy factors agree off the same halos. Within them each Gaussian shifted minimum energy is bounded by a fixed constant: try the local target lift, whose gradient energy is bounded and whose observation error is zero. Likewise, each isolated regular tile has shifted-to-centered ratio bounded below by a fixed positive constant after refinement. Theorem 3.1, including summation of constants and the uniform fractional mean, gives convergence to its Gaussian ratio; the same target trial bounds its Gaussian energy. These constants are uniform for large R once the microscopic mesh has been chosen. The exterior alone is unshifted. These facts give the remaining shift accounting in (4.5), even when an affected connected component has no global lift.

Cutting with prescribed guards. Cut disjoint larger boxes of tiles around the marks, well outside their guards. For joint partition weights $J_G(s, u)$, normalized at centered shifts and zero prescribed guard data, affine comparison gives

$$\frac{J_G(s, u)}{J_G(0, 0)} \leq \frac{J_{G^{\text{cut}}}(s, u)}{J_{G^{\text{cut}}}(0, 0)}. \quad (4.7)$$

This includes the connection, observation shifts, and prescribed primary heights as affine coordinates. Restoring the cut bonds is an increase of precision, so Lemma 2.2 applies. This is the comparison of [12, Equation (9.20)], now using the established Skellam comparison.

To return from zero prescribed data to unprescribed data, the remaining centered factor is a ratio of probabilities that all these guards are zero. Impose them successively. For each guard use a still larger separating patch, disjoint from the cuts and the previously imposed guards. Its centered pin probability in the full and clipped graphs lies between common open-patch and outer-pinned values, whose ratio is bounded by a separated hard-pin interaction. Their product costs at most e^{Cj} . Skip distances were chosen large enough to keep these boxes disjoint.

We may enlarge the affected region in (4.5) to include every otherwise untouched isolated tile in these boxes; their shifted ratios cost at most another e^{Cj} . Include the exterior object whenever it is met. Sum (4.7) over the data, keeping the same δ_i , since cutting beyond a guard does not change its interior kernel. The outside shifted-to-centered factor is at most one. Inside, the cut boxes are independent and their own shifted-to-centered factors are also at most one. All twists within a box are now ordinary bounded observation shifts after integer gauge changes. It remains to prove, in either of its two positive topologies,

$$\mathbb{E}_{\text{box}, s} \delta_i \leq e^{-cR}. \quad (4.8)$$

The local gain at typical data. Write $K_t(y, u)$, $t = o, c$, in a common bounded lift, where $y = Bv$, and cancel factors depending only on the guard or its exterior. Put

$$R(u) = \frac{K_o(0, u)}{K_c(0, u)}, \quad r_\Delta(u) = \frac{R(u)}{R(0)}, \quad F_t(y, u) = \log \mathbb{E}_{t,0}(e^{\langle y, Bh \rangle} \mid h_p = u).$$

The common observation constant $e^{-\|y\|^2/2}$ cancels as well. If

$$P_g(y) = \log \frac{Z_{g,U_o}(y)/Z_{g,U_o}(0)}{Z_{g,U_c}(y)/Z_{g,U_c}(0)},$$

the exact decomposition is

$$\log \frac{K_o(y, u)}{D_i^s K_c(y, u)} = \log \frac{R(0)}{D_i^0} + \log r_\Delta(u) + F_o(y, u) - F_c(y, u) - P_g(y). \quad (4.9)$$

The first term is $O(e^{-cR})$ in the refinement limit by the centered sandwich. The full-core pin ratio r_H of the *cut centered law* satisfies $\log r_H \leq \log r_\Delta \leq 0$, with H containing the restored bonds. No separate scaling limit for r_Δ is required: this sandwich and the bounded discrepancy δ_i suffice.

Theorems 3.1 and 3.2 give joint convergence of r_H and the conditional Laplace terms in the finite-observation sense, at this fixed geometry. They apply under the centered or tilted laws of either topology. One can first analyze conditioning in the whole cut box: its interior kernel given the full guard agrees with that in any completion with the same interior and guard. Transfer between cut and restored typical data follows explicitly as follows. For their centered guard laws μ_p, ν_p , Lemma 3.9 gives

$$r_H \leq \frac{d\nu_p}{d\mu_p} \leq e^{D(H,p)}.$$

Thus $\mu_p(E) \rightarrow 0$ implies $\nu_p(E) \rightarrow 0$, and

$$\mu_p(E) \leq \mu_p(r_H \leq \kappa) + \kappa^{-1} \nu_p(E).$$

Strict positivity of the limiting Gaussian r_H , followed by $\kappa \downarrow 0$, gives the converse. Taking a common finite list of probes therefore makes both conditional Laplace approximations and the pin-ratio approximation joint under either law. Uniform fixed-tilt exponential moments transfer exceptional sets to either tilted law by Hölder. Probe exhaustion is after refinement at this fixed geometry. Nothing here asserts a limit uniform at arbitrary microscopic guard values.

For completeness, the quantitative Gaussian calculation is as follows. Let u_c be the cut harmonic prediction from the guard in the centered massive energy, and let χ equal one on the core and vanish before the guard. With T denoting the gradient-plus-observation energy and $y = Bv$, Gaussian localization gives

$$\mathbb{E} \|\chi u_c\|_T^2 \leq \text{poly}(R) e^{-cR} (1 + \|y\|^2). \quad (4.10)$$

This is the localized-operator and Hilbert–Schmidt estimate in [12, Proposition 3.14], or [12, Equation (9.23)]. Restoration decreases centered guard covariance, and the deterministic prediction due to the tilt is bounded by the same localized operator. The Gaussian full-core pin ratio satisfies

$$-2 \log r_H^g \leq \|\chi u_c\|_T^2.$$

Indeed, subtracting this cutoff prediction makes the conditional mean zero on the core without changing its guard values. Orthogonal projection onto the restored admissible space also bounds the change of prediction between cut and restored topologies by $\|\chi u_c\|_T$. This remains valid when restoration matches two previously free traces: vanishing on the core is an admissible correction. Thus the linear-in-data part of the conditional Laplace discrepancy is bounded by $C\|y\|\|\chi u_c\|_T$. In fact, for $f = B^*y$, centered Gaussian conditional means m_t , and conditional covariances C_t^p ,

$$F_o(y, u) - F_c(y, u) = \langle f, m_o(u) - m_c(u) \rangle + \frac{1}{2} \langle f, (C_o^p - C_c^p) f \rangle.$$

Here $u_c = m_c(u)$ is always the cut-law prediction, evaluated under either topology's data, and $\|f\|_{T^*} \leq \|y\|$.

The conditional quadratic discrepancy from the open-patch comparison is at most

$$\text{poly}(R)e^{-cR}\|y\|^2.$$

One way to see all of its locality requirements is to compare the changes of the minimized observation-energy matrices under restoration, with and without the zero guard pin. Each change decays exponentially as an observation index recedes from the core, by positivity of the added form and a localized trial cut off near restored edges. Near the core, the guard pin's own effect decays from the guard. Cross entries follow by positive-form Cauchy–Schwarz. The same argument at the patch boundary compares the open patch with the full box. There are only polynomially many entries; guard-only and exterior terms cancel from the kernel ratio. Finite guard probes can be exhausted after these estimates. This is the projection argument underlying [12, Lemma 9.4].

Here $\|y\| \leq \text{poly}(R)$, so all these bounds have an exponentially small right side after weakening c . Use

$$\left| \frac{x-1}{x+1} \right| \leq \min\{1, |\log x|\}, \quad x > 0.$$

Convergence in probability of the finite-observation approximations, the bound $\delta_i \leq 1$, and Gaussian localization prove (4.8), with slack. At fixed R , increase m_* so it holds uniformly over the finite geometries and compact target families. Applying it in the disjoint boxes, absorbing the factors $2^j e^{Cj}$, proves (4.4). $\square$

4.3 Factorization, logarithmic errors, and rim insertions

The pattern bound alone must be supplemented by exact dependence information. Enlarge each mark halo past all skip and proxy radii, and join two potential marks if their enlarged halos meet or if they can attach to the same exterior terminal node. This defines a graph of bounded degree, independent of R, n, L . For disconnected groups of marks the normalized weights factor. Indeed, a skip depends only on nearby earlier marks, a considered proxy only on the current cuts in its patch, and an unmarked deletion isolates its tile. All final non-product integrals lie inside halos, possibly attached to the exterior. Distinct groups have disjoint variables and no surviving bonds between them. If the exterior could connect them, they were joined through its terminal node. The signed sums over final positive topologies therefore factor as well, with the original global order merely restricted to each group. All untouched factors cancel exactly. Violations of the skip rule have weight zero, a condition checked separately in each component.

Consequently the normalized partition is a finite gas of compatible connected mark sets, with activities bounded by e^{-cRj} for size j . We record explicitly the smallness argument for signed activities. In a graph of bounded degree, there are at most D^j connected sets of size j containing a given site, and their closed neighborhoods have at most $D'j$ sites. Let $\Xi(V)$ be the gas partition sum when marks are allowed only in a set V ; ordinary unmarked steps are still retained. Removing one allowed site x and summing over the connected polymer containing it gives

$$\frac{\Xi(V)}{\Xi(V \setminus \{x\})} - 1 = \sum_{\substack{P \ni x \\ P \subset V}} w(P) \frac{\Xi(V \setminus N[P])}{\Xi(V \setminus \{x\})}.$$

Induct on $|V|$. If smaller deletion ratios lie in $[1 - \varepsilon_R, 1 + \varepsilon_R]$, the absolute value of the right side is at most

$$\sum_{j \geq 1} D^j e^{-cRj} (1 - \varepsilon_R)^{-D'j}.$$

For sufficiently large R , this is at most $\varepsilon_R = O(e^{-c'R}) < 1/2$, after adjusting constants. The induction gives nonvanishing and positivity, also for all restricted sums, and

$$|\log \Xi(V)| \leq 2\varepsilon_R |V|. \quad (4.11)$$

Thus signed cancellations cannot invalidate the logarithmic estimate.

The shifted-to-centered ratios of corresponding no-mark proxies are identical in the lattice and Gaussian models, by (4.3). For final isolated tiles their ratios have arbitrarily small logarithmic discrepancies after refinement, by Theorem 3.1 and the uniform fractional mean in Theorem 2.5. Their Gaussian energies are uniformly bounded by trying the target; hence the limiting ratios are bounded away from zero. The isolated exterior has ratio one. Apply (4.11) to the two sectors of both models. Since there are at most $C_M(1 + \log n)$ processed tiles, this proves (4.2), choosing all local refinement errors smaller than a prescribed quantity tending to zero with R .

We next prove the tightness assertion of Proposition 4.2. For a primary inner-rim site x , insert e^{h_x} or e^{-h_x} in its bounded cut coordinates. The microscopic volume of every fixed-radius patch near x is bounded in terms of m_*, R, M , also for the rim translated to the other source. Add one distinguished item at its tile in the preceding expansion. Discard difference gains in a fixed neighborhood of this item; if an insertion lies in a marked kernel, first bound the absolute difference by the sum of its positive terms.

Clip a separate box about the distinguished item, including all necessary neighboring tiles, while retaining disjoint comparison boxes for more distant marks. Only boundedly many ordinary gains have been discarded. The prescribed-guard cutting inequality remains valid with the fixed real linear tilt $\pm h_x$ in the numerator, by Lemma 2.2. This uses monotonicity of the tilted ratio, not the bound one for it. In the distinguished box the ratio with the insertion is bounded by a finite constant: each component has confining average penalties, there are boundedly many lattice sites, the edge sums have all fixed exponential moments, and translation sums are Gaussian-confined. This constant may depend arbitrarily on the fixed m_*, R, M . The inserted functional remains literally $\pm h_x$ throughout this comparison, not an extension by a flow or a harmonic prediction. After cutting it is supported entirely in the distinguished box. No matching constraint remains to its exterior, so every other factor is untilted and has shifted-to-centered ratio at most one. A bounded change of chart contributes only a bounded scalar. The rounded distinguished graphs form a finite set at fixed m_* , and their target vectors form compact sets, making the finite insertion constant uniform for all integer n .

The previous centered accounting applies with the distinguished halo included. Connected weights containing the distinguished item are therefore bounded by

$$C_{m_*, R, M} e^{-cRj},$$

where j is the number of ordinary marks; the bounded number of lost gains is absorbed in the constant. All other components factor as before. More explicitly, let $*$ be the forced distinguished item and let Ξ be the gas without the insertion. Group the inserted expansion by the connected component containing $*$. With $P \subset V$ its ordinary marks, the exact normalized expression is

$$\mathbb{E}_s e^{\pm h_x} = \sum_{\substack{P \subset V \\ P \cup \{*\} \text{ connected}}} W_*(P) \frac{\Xi(V \setminus N[P \cup \{*\}])}{\Xi(V)}.$$

The weights include the finitely many altered factors in the distinguished halo. For $j = |P|$, their absolute values are at most $C_{m_*, R, M} e^{-cRj}$, the number of connected choices is at most D^{j+1} , and

the removed neighborhood has at most $D'(j+1)$ sites. The deletion bounds proving (4.11) therefore dominate this expression by

$$C_{m_*,R,M} \sum_{j \geq 0} D^{j+1} e^{-cRj} (1 - \varepsilon_R)^{-D'(j+1)} < \infty.$$

This bound is independent of $|V|$; no extensive log-partition bound is used when dividing the two sums. Consequently,

$$\mathbb{E}_s e^{h_x} + \mathbb{E}_s e^{-h_x} \leq C_{m_*,R,M}, \quad (4.12)$$

uniformly in n, L . There are boundedly many inner-rim sites for fixed m_* , so (4.12) proves the asserted absolute tightness and completes the proof of Proposition 4.2.

4.4 Filling the holes and the angular energy

Return to the full dual partition of Lemma 2.1. First add the observation penalties, only outside the holes; this lowers its shifted-to-centered ratio. Compare this weighted full partition with its punctured version. With normalized edge weights p_b , the centered filling kernel is bounded above by one: retain a spanning forest attaching the finitely many removed vertices to their rim, sum the forest increments using $\sum_j p_b(j) = 1$, and bound the other edge factors by one. A fixed harmless normalization convention would instead give a fixed upper bound.

By Proposition 4.2, in the shifted punctured law all adjacent rim values are bounded by a fixed constant with uniformly positive probability. Fill both holes with any fixed bounded configuration in the cut coordinates. There are only finitely many required integer differences, bounded in terms of the fixed choices, including those on the continued offset path. All their weights $p_b(j)$ are strictly positive. Thus the numerator filling factor is bounded below on this event by a positive constant. We obtain

$$\mathbb{E}_{\mu_{L,b}} \cos(\theta_0 - \theta_{ne_1}) \geq c_{m_*,R,M,b} \frac{Z^s}{Z^0} \quad (4.13)$$

uniformly for the large outer boxes under consideration. The exterior dual face remains fixed at zero throughout, which is the original free spin boundary condition; no primal boundary spin has been fixed.

For the real Gaussian model, completing the square gives the shifted-to-centered ratio as the exponential of minus one half the minimum affine energy. Trying the sampled target v makes the observation error zero and matches the exterior pin. Therefore

$$-\log \frac{Z_g^s}{Z_g^0} \leq \frac{1}{2a} (4\pi \log n + O_{m_*}(1)). \quad (4.14)$$

Indeed, near each source the angular energy is

$$\int_{m_*/n < |z| < c} |\nabla \arg z|^2 \, dz = 2\pi \log n + O_{m_*}(1).$$

Replacing the circular inner boundary by a square changes only the bounded term. The other angle and the cutoff corrections are smooth near that source; their cross terms with its $1/|z|$ gradient are integrable. All remaining energies are bounded. On the lattice, derivative errors of order r^{-2} at distance r give summable errors in the squared-gradient sum. Thus the two singularities give exactly 4π , with no extra factor from oriented edge counting.

Proof of Proposition 4.1. Choose M , then arbitrarily large fixed R , and then sufficiently large fixed m_* as in Proposition 4.2. Combining (4.2), (4.13), and (4.14) gives, uniformly in sufficiently distant outer boundaries,

$$\log \mathbb{E}_{\mu_{L,b}} \cos(\theta_0 - \theta_{ne_1}) \geq -\frac{2\pi}{a} \log n - C_M(1 + \log n)\eta_R - O_{m_*,R,M,b}(1).$$

For each fixed n , first take the prescribed free-box limit $L \rightarrow \infty$. Divide by $\log n$, and then take the liminf as $n \rightarrow \infty$. All hole and insertion constants were fixed before this limit and disappear. Finally let $R \rightarrow \infty$, obtaining the claimed lower exponent. Since $0 < C_b(n) \leq 1$, this polynomial lower bound also implies $m(b) = 0$. $\square$

5 Openness above the marginal coefficient

We now exclude a supermarginal coefficient at the mass threshold. The required assertion concerns the original Skellam height law:

Proposition 5.1. *If $a(b) > 8\pi$, there is an open interval I containing b , contained in $(0, \infty)$, such that $a(\tilde{b}) > 0$ for every $\tilde{b} \in I$.*

We will not assume continuity of a , nor differentiate it. The argument first adds a positive precision on *block averages*. This supplies a Gaussian large-field bound for the observed field, while its scaling coefficient can be kept above 8π . We prove entry into the small-activity maps at fixed reference, and then use a torus variance diagnostic to return to the original model.

5.1 A strengthened law and its local limits

Fix b with $a = a(b) > 8\pi$. Choose $\delta > 0$ sufficiently small that

$$a' = (a^{-1} + \delta)^{-1} > 8\pi, \quad \alpha = \sqrt{a'}. \quad (5.1)$$

Both δ and a' remain fixed when the physical parameter $\tilde{b}$ varies. For a dyadic s_0 , let B_{s_0} average over disjoint square cells of side s_0 , and let A_{cell} be the unit-conductance nearest-neighbor Laplacian on these cells. Add to the height energy the term

$$\frac{\delta}{2} \langle B_{s_0} h, A_{\text{cell}} B_{s_0} h \rangle. \quad (5.2)$$

All grids are nested and have the common square-symmetry origins used below. Set $s = ts_0$, where t is also dyadic, and observe

$$Y = B_s h + \eta, \quad \eta \sim N(0, \text{Id}), \quad (5.3)$$

with independent observation noise. On tori we work modulo global height translation by 2π . When local observations confine a component constant, its translations are summed instead.

The comparison observation is $B_s \alpha \psi + \eta$, where the mean-free covariance of ψ is A^+ and its constant is uniform modulo $2\pi/\alpha$. In particular, this is the *ordinary nearest-neighbor* Gaussian reference; no averaged precision is added to it. On a cut graph G , write its minimized observation energy as

$$E_G(y) = \inf_v \left\{ a'^{-1} \langle v, A_G v \rangle + \|B_s v - y\|_2^2 \right\}, \quad (5.4)$$

using free constants on the components. Denote by $Z_G(y; \tilde{b})$ the corresponding Skellam partition sum with (5.2) and observation penalty $\|B_s h - y\|_2^2/2$.

The local domains are the mixed polygonal patches of Theorem 3.1, aligned with whole s -cells. Whenever microscopic bonds between tiles are deleted, delete also the bonds of A_{cell} crossing that cut. Thus a pre-pin component has precisely its own subcells and face bonds. A hard core pin used to dominate a restoration includes all primary vertices in every subcell supporting a restored averaged bond. Such supports fit inside the fixed core collars.

Lemma 5.2 (Strengthened local limits). *Fix a bounded mixed geometry in observation-cell units, with separated positive-area core and guard pins and observation masses controlling each unpinned constant. At the central parameter b , the strengthened laws have the Gaussian smear, conditional-smear, and pin-interaction limits of Theorems 3.1 and 3.2, with coefficient a' , in the iterated limit*

$$\lim_{t \rightarrow \infty} \lim_{s_0 \rightarrow \infty} .$$

The limits hold under each indicated pin set. The quantities D and r_H retain their centered-law definitions; finite-probe convergence of these conditional quantities holds also when their data are sampled under any fixed real observation tilt. This is convergence in probability under the corresponding guard laws, not a redefinition of the centered pin ratios. At fixed geometry the conclusions are uniform on compact sets of observation data.

Proof. For fixed t , the additional energy is a finite-rank quadratic tilt of finitely many subcell averages. Theorems 3.1 and 3.2 give its limit on each fixed pin set. Integration against the nonnegative quadratic weight is justified by its boundedness and the confining observation masses; fixed linear tilts are handled by the exponential moment bounds. Free component constants are treated by the joint uniform-phase limit before summing their confining translations.

The full pin interaction also transforms by finite-dimensional quantities. If W is the added quadratic weight, and subscripts indicate centered conditioning on the designated pins, then

$$D_W(H, p) - D(H, p) = \log \mathbb{E}_{H,p} W + \log \mathbb{E} W - \log \mathbb{E}_H W - \log \mathbb{E}_p W. \quad (5.5)$$

Each expectation has a strictly positive Gaussian limit. This proves, at fixed t , convergence to the pin interaction for the tilted continuum Gaussian precision, not merely convergence of unconditioned smears.

In observation-cell units, let P_t denote averages over squares of side $1/t$, and B averages over the unit observation cells. The continuum form obtained in this first limit is

$$T_t(f) = a^{-1} \int |\nabla f|^2 + \delta \sum_{\langle i,j \rangle} ((P_t f)_i - (P_t f)_j)^2 + \|Bf\|_2^2. \quad (5.6)$$

It converges in inverse forms to

$$T(f) = a'^{-1} \int |\nabla f|^2 + \|Bf\|_2^2, \quad (5.7)$$

also with any of the specified zero-region pins. Here is the form argument, including its normalization. If $\zeta = 1/t$, the averaged-gradient term is a sum of $\zeta^2 |D_\zeta P_t f|^2$. Bounded T_t -energy gives weak H^1 and strong L^2 compactness on each fixed component. The averaged difference quotients converge weakly to the directional derivatives on its interior, giving the lower bound with coefficient δ . Conversely, smooth functions give the matching upper bound by Riemann sums. The averaged-gradient term is uniformly bounded by a constant times continuum gradient energy, by the two-cell Poincare inequality. Energy approximation therefore extends recovery to all admissible functions, using corner cutoffs and density with zero neighborhoods of the pins, as in [12, Lemma 3.12]. No component

is glued through a point. This proves the claimed inverse-form convergence and, in particular, convergence of all finite covariance and pinned inverse matrices.

We also need convergence of the infinite-dimensional interaction determinant. Form convergence alone would not give it. We verify the uniform compactness used in [12, Proposition 3.14] for the additional averaged term. For an energy-harmonic function off a pin or source, test its equation against $\chi^2 f$, where χ is a smooth cutoff in a separating margin. On a subcell replace $(P_t(\chi^2 f))_i$ by $\chi_i^2(P_t f)_i$. The error satisfies

$$\left| (P_t(\chi^2 f))_i - \chi_i^2(P_t f)_i \right| \leq C(t^{-1}|\chi_i| + t^{-2})(P_t f^2)_i^{1/2}. \quad (5.8)$$

Pairing this error with each incident averaged bond and applying Cauchy–Schwarz absorbs the term with $|\chi_i|$ into a fixed part of the positive weighted bond energy. The remaining t^{-2} terms cost local L^2 mass: use

$$|(P_t f)_i - (P_t f)_j| \leq (P_t f^2)_i^{1/2} + (P_t f^2)_j^{1/2}$$

and the bounded number of incident bonds. The differences of the cutoff values themselves are controlled by the discrete product inequality. All bounds are uniform for large t .

The unit-cell observation term may cross an intermediate cutoff. On the fixed geometry its forcing consists of finitely many bounded cell-average coefficients. Carry these coefficients as finite-dimensional coordinates through the successive restriction maps. Their bounds follow from the original energy bound. The remaining terms are local on an enlarged margin for large t . We consequently obtain the same nested gradient-from- L^2 estimates as for the form with unit-cell mass alone, with this fixed finite-rank enlargement.

Two broken-domain H^1 -to- L^2 compact embeddings, each with singular numbers $O(j^{-1/2})$, separated by these harmonic interior estimates, give a uniform $O(j^{-1})$ singular-value tail for the cross projections between the separated pin regions. A fixed finite-rank enlargement changes only finitely many singular directions. Bounded-energy sequences harmonic off one pin have weak limits harmonic off that pin for T : test their equations against smooth functions, whose averaged differences converge strongly. For a weakly null sequence, compact L^2 convergence and the interior estimate give strongly vanishing energy near the other pin; the observation averages vanish as well.

For completeness, identify all the varying energy spaces with the same broken H^1 space V . The forms T_t are uniformly equivalent there, since the original gradient and observation terms are a lower bound and the averaged-gradient term has a uniform upper bound. Let $V_{0,H}$ be the fixed subspace of functions zero on H . Its T_t -orthogonal complement U_H^t is the source projection space. If ϕ is a smooth density supported in H , orthogonality of $v_t \in U_H^t$ to the T_t -Riesz representative of ϕ is exactly $\int v_t \phi = 0$, independently of t .

Take T_t -unit vectors in U_H^t orthogonal to increasing finite lists of these probes. Every weak limit vanishes on H and is T -harmonic off H , by the smooth-test passage above. It lies both in $V_{0,H}$ and its T -orthogonal complement, so is zero. The finite observation forcing coordinates are the actual averages Bv_t ; strong L^2 convergence makes them zero too. For a cutoff χ_p equal to one near the opposite pin, the local estimate gives $T_t(\chi_p v_t) \rightarrow 0$. Moreover $\|P_p^t v_t\|_{T_t} \leq \|\chi_p v_t\|_{T_t}$, since $v_t - \chi_p v_t$ vanishes on p . This proves uniform finite-probe approximation of the cross projection. The same argument applies in reverse, and the finite-probe Gram matrices converge by inverse-form convergence. Combined with the uniform square-summable singular tails, these facts give convergence of the cross determinants. The limiting cross projection has norm strictly smaller than one, as in [12, Proposition 3.14]; operator convergence preserves a norm gap. In particular the logarithmic determinant defining $D(H, p)$ converges.

The conditional conclusions now follow by the regression and minimal-Hessian argument used in Theorem 3.2, with the covariance and determinant just obtained. Conditional Laplace transforms

are first approximated using finitely many smooth guard probes. The full pin ratio follows by increasing finite core penalties and using the centered identity $\mathbb{E}r_H = e^{-D(H,p)}$. For use in both neighboring restoration topologies, take a common finite probe list. The bound $r_H \leq r_\Delta \leq 1$ from Lemma 3.9, and strict positivity of the Gaussian limiting r_H , transfer negligible exceptional sets in both directions. Fixed real tilts preserve convergence for these same centered-defined quantities by the exponential moment bounds; the centered expectation identity is not asserted under tilted sampling.

Every geometry and probe list is fixed before refinement. Compact-data uniformity follows by taking convergent sequences of observation vectors and using the same exponential integrability. Operationally, given an error tolerance one first chooses a sufficiently large t , then a sufficiently large s_0 depending on t . No single refinement threshold uniform over all t is required. $\square$

5.2 A refinement-independent large-field reserve

The bare quadratic energy of the subdivided Skellam law cannot be used as a uniform site-level Gaussian tail. The next bound is the reason for adding (5.2).

Lemma 5.3. *For every cut topology used in the entry construction, there is $c_* > 0$, depending only on δ, a' and the fixed local geometry conventions, such that*

$$\frac{Z_G(y; \tilde{b})}{Z_G(0; \tilde{b})} \leq \exp\{-c_* E_G(y)/2\}. \quad (5.9)$$

The same constant works for all sufficiently large dyadic refinements and for $\tilde{b}$ in a neighborhood of b .

Proof. On the subdivided quadratic lattice, complete the square in the joint height and observation variables. Centered theta domination bounds the shifted sum by the centered sum times the exponential of minus half its real minimum energy. Discard the nonnegative microscopic chain energy in that minimum, retaining the added average energy and $\|B_s h - y\|_2^2$. We show that this retained energy dominates $c_* E_G(y)$ for every real trial height.

Write $u_i = (B_{s_0} h)_i$. On the union of two adjacent t -by- t subcell squares, the ordinary grid Poincare inequality gives

$$|\bar{u}_{\text{left}} - \bar{u}_{\text{right}}|^2 \leq \frac{2}{t^2} \sum_i |u_i - \bar{u}|^2 \leq C \sum_{\langle i, j \rangle} (u_i - u_j)^2. \quad (5.10)$$

The constant does not depend on t . Apply this to each same-component observation face and sum with bounded overlap. The observation residuals then show that

$$\sum_{\langle x, x' \rangle \text{ in } G} (y_x - y_{x'})^2 \leq C_\delta \left\{ \delta \sum_{\langle i, j \rangle} (u_i - u_j)^2 + \|B_s h - y\|_2^2 \right\}. \quad (5.11)$$

Only retained faces within a component occur.

Conversely, $E_G(y)$ is at most a constant times the left side of (5.11). One explicit trial interpolates the values y_x at scale s , with central plateaus and a piecewise affine subdivision joining cell centers, face midpoints, and vertices. At a midpoint or vertex use the local average within each incident face-connected sector. Both its gradient energy and its observation error are bounded by the nearby face-difference sum. There is no bond requirement between opposite-only sectors. These local

estimates are uniform under the permitted cuts, and agree under lattice refinement. They prove the required comparison of minima.

The constants do not use the microscopic chain precision or the value of $\tilde{b}$. Pass to the fixed-graph Skellam limit using Lemma 2.2. If necessary decrease c_* so that $c_* \leq 1$. This gives (5.9) uniformly in the stated parameters. $\square$

5.3 Entry into the full analytic activity norm

We use the nearest-neighbor Gaussian split and local subset algebra of [12, Theorem 9.1]. The theorem there is stated for quadratic discrete heights; it is *not* being applied directly to the present physical law. We prove its needed entry conclusion using Lemmas 5.2 and 5.3, while retaining its model-independent Gaussian estimates and finite expansion.

Fix admissible downstream parameters for [12, Proposition 8.3] in the gapped regime $a' > 8\pi$. Write L for the block factor, $\mathfrak{A}$ for the activity weight, and h_0 for the analytic radius. Choose with strict margins

$$\max(0, 1 - c_*) < h_1 < h_* < 1/p_2, \quad 1 < p_1 < p_2. \quad (5.12)$$

The observation-energy regulator and norm are precisely those of [12, Equation (9.13)] and [12, Equation (9.3)], for this fixed nearest-neighbor reference and these parameters. For orientation, an activity $K(X, \psi)$ is a local analytic correction indexed by a finite set X of blocks; $|X|$ is its block count and ψ is the reference field. The point norm $|K(X)|_{u, X, \psi}$ includes the full sum of field derivatives at analytic radius h_0 . At block side $u = mL^j$, the global norm is

$$\|K\|_{u, \mathfrak{A}, V} = \sup_{X, \psi} \mathfrak{A}^{|X|} \frac{|K(X)|_{u, X, \psi}}{W_u^\kappa(X, \psi) e^{V_j(X, \psi)}}.$$

The block supports and ordinary gradient regulator W_u^κ are those of [12, Lemma 4.3] and [12, Lemma 4.4]. The observation-energy regulator V_j is displayed in (5.23) below. Thus the entry estimate controls all fields and analytic derivatives, not only bounded real data.

Lemma 5.4 (Entry for the strengthened law). *For every $\epsilon > 0$ there are fixed dyadic integers q, R, t, s_0 , with $P = R^5$, $s = ts_0$, and $m = qRs$, and a real interval I about b , such that the exact relative observation density of the strengthened law, after averaging the observation noise and reference modes below scale m , has plane and torus local subset activities K_m satisfying*

$$\|K_m(\tilde{b})\|_{m, \mathfrak{A}, V} \leq \epsilon \quad (\tilde{b} \in I). \quad (5.13)$$

The norm includes the full analytic field-derivative sum. The activities are real, even, square-covariant, invariant under block-grid translations and common field translations by $2\pi/\alpha$. The torus representation is exact for compatible sides $n \geq sP$, with exact plane copying on unwrapped padded supports and the copying property under subsequent local maps.

Proof. We give the entry estimates, especially the places where the physical law differs from the quadratic source. The choices are made in this order: the downstream norm and regulator parameters and imaginary strip widths; q, R, P ; a finite large-field threshold; t ; s_0 depending on t ; and finally the interval I .

The reference split and exact expansion. For this observed law the exact density relative to its Gaussian reference is, up to normalization,

$$Z_{\text{full}}(y; \tilde{b}) \exp\{E_{\text{full}}(y)/2\}. \quad (5.14)$$

This is the observation identity [12, Equation (9.2)] with the new Z . Use the positive finite-range nearest-neighbor covariance $C_{<m}$ of [12, Equation (9.11)], followed by its positive finite-range shells. These covariances, their ranges, and the reference energies do not involve (5.2).

Use the covariant isolation of [12, Section 9.2]: tiles of diameter comparable to R observation cells, with trimmed vertex tiles, finite colors of a fixed period q , and bounded-size proximity groups within each color. The core, open patch, and surrounding guard have fixed multiples of R as their widths. At an event, delete all remaining bonds connecting its tiles to other tiles, counting each bond once and deleting the corresponding averaged bonds. If o, c are the current open patches before and after deletion, use the replacement

$$\mathcal{B}_i(y) = C_i \exp\{-(E_o(y) - E_c(y))/2\}, \quad C_i = \frac{Z_o(0; b)}{Z_c(0; b)}. \quad (5.15)$$

The scalar is fixed at b , while the Gaussian shape always has reference a' . Replace the old weight by this proxy times the cut weight plus their difference. A selected difference marks the event and skips all later events within a fixed sufficiently large halo. This is an exact finite expansion. A later considered patch avoids each earlier marked core and guard, so its proxy is independent of the sign chosen inside that difference.

Divide by the product of the no-mark proxies and centered isolated-tile normalizers. Outside mark halos expand the isolated error

$$\frac{Z_Q(y; \tilde{b}) e^{E_Q(y)/2}}{Z_Q(0; b)} - 1.$$

The residual Gaussian quadratic difference between the full energy and the no-mark energy is expanded into links, as in [12, Equation (9.9)] and [12, Equation (9.10)]. Gaussian localization gives link weights $Ce^{-c(R+d)}$ for length d , with their full displacement retained on tori. All averaged-bond supports lie inside the required collars; they change the physical positive topologies, not these Gaussian link estimates.

Local normalization and cutting. For a marked core, its centered restoration ratio is sandwiched between the open-patch ratio and the ratio with its outer guard pinned to zero. The logarithmic gap is bounded by $D(H, p)$, with H containing every restored support. This follows from Lemma 3.9 also for the additional positive averaged form. Lemma 5.2 and Gaussian localization make this gap exponentially small in R , after the two refinements at the fixed geometry.

Only $O(j_1)$ centered comparisons are charged for j_1 marks. Indeed telescope separately the positive branch and no-mark branch on an enlarged union of their halos. The two paths agree at its artificial boundary, so clipped proxies there cancel; final integrals and proxies outside the affected union cancel as well. All events remaining inside the union number $O(j_1)$. This is the finite double telescoping of [12, Lemma 9.3]; it applies unchanged to the present centered sandwich. Finite-patch continuity preserves the bounded comparisons near b , with the constants C_i fixed as in (5.15). No error proportional to the unaffected volume is introduced.

For a positive topology G , the resulting bound has the form

$$e^{Cj_1 + Ce^{-cR}\mathcal{E}(y)} e^{E_G(y)/2} \frac{Z_G(y; \tilde{b})}{Z_G(0; \tilde{b})} \mathbb{E}_{G,y} \prod_i \delta_i, \quad (5.16)$$

where $\mathcal{E}$ is a face-difference cost in the padded halo and, at prescribed primary guard data γ_i ,

$$\delta_i = \frac{|K_o(y, \gamma_i) - \mathcal{B}_i(y) K_c(y, \gamma_i)|}{K_o(y, \gamma_i) + \mathcal{B}_i(y) K_c(y, \gamma_i)} \leq 1.$$

All factors exterior to a full guard cancel in this ratio.

Cut into P -boxes on an optimized translate of the R -grid. Averaging translates loses at most $C(R/P)j_1$ marks and costs at most $C(R/P)\mathcal{E}(y)$ in Gaussian energy. The joint shifted-versus-centered comparison at prescribed guard data is Lemma 2.2, with the observations embedded as height-plus-noise coordinates. Its remaining centered guard factors cost e^{Cj_1} : impose the guards sequentially and use open and zero-guard bounds with separated outer collars. Thus the cut comparison has the same seam charge as [12, Equation (9.21)],

$$e^{Cj_1+C(R/P)\mathcal{E}(y)}. \quad (5.17)$$

Bounded data and large data. We distinguish the graph used to classify observation data from the positive topologies being compared. Let D^0 be the observation-cell graph on the pattern's original padded support D , *before any isolation deletion*. Clip this graph by the P -box grid and put

$$\mathcal{E}_{\text{box}}(y) = \sum_{\substack{\langle x, x' \rangle \text{ in } D^0 \\ x, x' \text{ in the box}}} (y_x - y_{x'})^2. \quad (5.18)$$

In particular, a face remains in this cost when an isolation step deletes its physical bonds. Retain a deep mark only if its complete open patch, restored supports, and guard lie in the same clipped box; the other marks are among the boundary losses already charged in (5.17). Thus every face across which any retained restoration, Gaussian proxy, or guard comparison tests a relative translation belongs to this original graph. Clipping adds no observation cell or exterior field value to D .

Fix the large-field threshold temporarily. On each connected component of this *original* clipped graph translate all heights, observations, and guard data together by one height period, putting one observed value in $[-\pi, \pi]$. Every factor under comparison is invariant under this common translation. Bounded $\mathcal{E}_{\text{box}}$ now gives a compact set of observation values by following paths in the finite original graph. Independent translations of components created by a later cut are not used: a restoration may join those components and detect their relative translation. At fixed R, P there are finitely many geometries, positive topologies, and mark choices. Each Gaussian energy and each physical partition function still uses its own actual positive topology, not D^0 .

Decompose the logarithm of $K_o/(\mathcal{B}_i K_c)$ into its centered zero-guard value, the variation of its centered restoration with guard data, and its two conditional observation Laplace terms minus the Gaussian proxy change. The first term is $O(e^{-cR})$ after refinement. The second lies between $\log r_H$ and zero. The conditional terms and r_H converge by Lemma 5.2 under either topology and its fixed observation tilt.

In the limiting Gaussian energy, let u_c be the cut harmonic prediction from guard data and let χ_H equal one at the core and vanish before the guard. The unchanged nearest-neighbor localization estimate [12, Equation (9.23)] gives, for every $r \geq 2$,

$$\|\|\chi_H u_c\|_T\|_{L^r} \leq C\sqrt{r} \text{poly}(P)e^{-cR}(1 + \sqrt{\mathcal{E}_{\text{box}}(y)}). \quad (5.19)$$

The squared cutoff energy bounds $-2\log r_H$. Correcting a cut minimizer by this cutoff makes it admissible for restoration without changing the guard; orthogonal projection therefore bounds the linear conditional-Laplace discrepancy by $C\|y\|_2\|\chi_H u_c\|_T$. The conditional quadratic discrepancy from the open-patch proxy is bounded by $Ce^{-cR}\|y\|_2^2$, by the same guarded localization. These arguments use finite-energy predictions and finite guard probes, not pointwise continuum guard values.

Transfer these estimates using the conditional approximation and $r_H \leq r_\Delta \leq 1$. The Gaussian limiting r_H is strictly positive, so exceptional sets are negligible in both topologies, also after fixed

tilts. On such sets use $\delta_i \leq 1$. Since $|(x-1)/(x+1)| \leq \min(1, |\log x|)$, bounded moments of each δ_i have the bounds from (5.19). Holder over the at most $\text{poly}(P)$ deep marks, and absorption of polynomial losses, give

$$C \exp\{-cRj_{\text{deep}} + R^{-12}\mathcal{E}_{\text{box}}(y)\}. \quad (5.20)$$

For example, maximize $(1+\mathcal{E})^{Cj}e^{-R^{-12}\mathcal{E}}$. Since $j \leq \text{poly}(P)$, a factor $[P^C j^C (1+\mathcal{E})^C]^j$ is bounded by $e^{O(j \log R) + R^{-12}\mathcal{E}}$. The $O(j \log R)$ loss is absorbed by the exponential rarity. We use this smaller allowance so that the conservative P^2 coordinate count is negligible below. Isolated selected-tile errors can be made smaller than e^{-cR} by the same compact Gaussian convergence.

We also need a separate estimate after a difference is expanded into its two positive terms. For every actual positive box topology G , Lemma 5.2 gives

$$e^{E_G(y)/2} \frac{Z_G(y; b)}{Z_G(0; b)} \longrightarrow 1$$

uniformly on the good-data compact set just described, after the common period recentering. The reference limit is precisely the Gaussian of coefficient a' , on that same topology. There are only finitely many such topologies. Choose the refinements so that, on every good box,

$$e^{E_G(y)/2} \frac{Z_G(y; \tilde{b})}{Z_G(0; \tilde{b})} \leq 2 \exp\{R^{-12}\mathcal{E}_{\text{box}}(y)\}. \quad (5.21)$$

At $\tilde{b} = b$ choose the stronger upper bound $3/2$ using that convergence; finite-patch continuity gives the bound 2 after shrinking the eventual parameter interval. For an isolated-tile error expanded into its positive terms, the constant Gaussian term satisfies the same bound. Centered normalizations between branches cost the already proved e^{Cj_1} . Equation (5.21) therefore bounds positive branches without any discrepancy factor δ_i ; it is not obtained by removing δ_i from (5.20). The displayed allowance is retained to use the same quadratic-cost accounting in both estimates.

On a bad box discard rarity and apply Lemma 5.3. Its centered-normalized factor is at most

$$e^{E_G(y)/2} \frac{Z_G(y; \tilde{b})}{Z_G(0; \tilde{b})} \leq e^{(1-c_*)E_G(y)/2}.$$

This replaces the bare quadratic estimate [12, Equation (9.24)]. It is the only large-field input needed from the physical height law.

Gaussian averaging with reserve. Here is the energy accounting for the original-face costs. For any real trial field v on the uncut padded domain, two-cell Poincare and the observation residual give

$$\sum_{\langle x, x' \rangle \text{ in } D^0} (y_x - y_{x'})^2 \leq C\{a'^{-1}\langle v, A_D v \rangle + \|B_s v - y\|_2^2\}. \quad (5.22)$$

Indeed insert the two cell averages of v in each difference. The difference of averages is bounded by the gradient energy on the two adjacent cells, as in (5.10); the two residuals are charged at most four times. All faces are internal to D^0 , so these two-cell regions are present before minimization and have bounded overlap, also at a clipped boundary. Minimizing over v proves the bound by $CE_D(y)$. This argument does not ask a cut topology to control a deleted face.

For every positive branch of the expansion, restriction of that same uncut trial field to the disjoint cut boxes bounds the sum of their actual minimized energies E_G by E_D . The bad-box physical contribution is consequently at most $(1-c_*)E_D/2$. All remaining data costs—the $R^{-12}\mathcal{E}_{\text{box}}$

good-box allowance, seam and halo costs, Gaussian links, and $\varepsilon_1 \mathcal{E}_{\text{box}}$ bad-box threshold charges—use faces of D^0 . Their summed coefficients are at most $C(R^{-12} + R/P + e^{-cR} + \varepsilon_1)$ per face. For halos this follows from their bounded multiplicity; for links one sums their exponentially decaying displacement weights. The same bound applies to every restoration branch. Choose R large and then $\varepsilon_1 > 0$ small so that (5.22) makes their total less than $[h_1 - (1 - c_*)]E_D/2$. Thus the total positive quadratic cost is at most $h_1 E_D/2$, with strict reserve as required by (5.12). The large-field threshold is chosen only after this reserve is fixed.

The reference covariance contracts the observation energy, exactly as in [12, Equation (9.15)]. At scale $u = mL^j$, the regulator is

$$V_j(X, \psi) = \frac{1}{2} \sup_{\zeta, e} \left\{ h_* E_{D_u(X)}(B_s \alpha(\psi + \zeta) + e) - p_1^{-1} (\|\zeta\|_{<u}^2 + \|e\|_2^2) \right\}. \quad (5.23)$$

Here the first norm is the Cameron norm of the reference pieces below u ; singular covariances mean their Gaussian Hilbert spaces. The covariance contraction and $p_1 h_* < 1$ make this supremum finite. Its composition, localization, and terminal bounds are [12, Lemma 9.2]; these involve only the unchanged reference covariance and whole observation cells.

The determinant actually integrated is charged by its localized quadratic costs, not by the larger dominating form $h_1 E_D$. Its trace is bounded by

$$C \text{poly}(P) N_{\text{bad}} + CR^{-12} P^2 N_{\text{good}} + C(R/P) R^2 (j_1 + j_2) + C \sum_{\ell} e^{-c(R+d_{\ell})} \text{poly}(d_{\ell}). \quad (5.24)$$

Here N_{good} counts occupied good boxes. Broad bad-box terms, including their threshold charges, have rank $O(P^2)$. The good-box allowance in (5.20) contributes at most $CR^{-12} P^2$ per occupied box, counting all present cells of the clipped box. Here P^2 is only an upper bound on their number; the actual quadratic cost remains inside D . A halo intersects a bounded number of P -boxes, so $N_{\text{good}} \leq C(j_1 + j_2)$. The remaining non-broad halo terms have coefficient $O(R/P)$ on $O(R^2)$ observation cells, and links retain their exponentially small path costs. The added physical precision is contained inside Z_G ; it is not a term of this reference Gaussian cost matrix. Covariance contraction gives a spectral margin below one, so logarithmic determinant costs are bounded by a fixed multiple of this trace. No determinant counts subcells or microscopic height variables.

Insert on a bad box the upper bound

$$1 \leq \exp\{\varepsilon_1(\mathcal{E}_{\text{box}}(y) - U_{\text{big}}^2)\},$$

with sufficiently small fixed $\varepsilon_1 > 0$. Choose the finite U_{big} to pay the polynomial bad-box losses, the first term of (5.24), and the lost item gains. This choice precedes t, s_0 . Since $P = R^5$, both the full good-box allowance $R^{-12} P^2$ and the halo seam trace cost are $O(R^{-2})$ per item. Gaussian completion with reserve consequently gives, for an averaged pattern F_{π} ,

$$|F_{\pi}(\psi)| \leq C^{j_1+j_2} e^{-cR(j_1+j_2)} \prod_{\ell} C e^{-c(R+d_{\ell})} W_m^{\kappa_0}(X, \psi) e^{V_0(X, \psi)}, \quad (5.25)$$

where $\kappa_0 < \kappa$ leaves ordinary regulator reserve. A real translate by xf , with f of scaled point norm at most one, costs at most $e^{Cx^2|X|}$, with the final regulator still evaluated at ψ . This uses $h_1 < h_*$ in the optimized quadratic form and $\langle f, A_D f \rangle \leq C|X|$.

Imaginary directions and analytic derivatives. The full analytic norm cannot be inferred from bounded-real-data convergence alone. On the extended quadratic lattice, including (5.2), the

theta parity identity [12, Equation (9.29)] gives

$$\frac{|Z_G(y + iz; \tilde{b})|}{Z_G(y; \tilde{b})} \leq \frac{Z_G(iz; \tilde{b})}{Z_G(0; \tilde{b})}. \quad (5.26)$$

It survives the fixed-graph Skellam limit. The centered imaginary ratio is its Gaussian prefactor times a positive characteristic function; centered pinning increases that characteristic factor.

Expand differences into positive topologies, temporarily discarding their rarity, and apply (5.26) to each whole integrated topology. Pin strips of width $R/4$ along translated P -box walls. All walls and strip boundaries are on observation-cell boundaries: R is dyadic, $R/4$ is integral, and the translates use the R -grid. Thus strips are unions of whole observation cells and of their nested subcells; no observation average is split. Cutting through their interiors disconnects also the averaged bonds, because all their supporting subcells there are pinned. On the resulting bounded boxes, Lemma 5.2 gives the Gaussian imaginary ratios uniformly on bounded imaginary directions, with an arbitrarily small multiplicative error after refinement. Their limits are strictly positive on each such compact set.

The additional Gaussian wall energy, averaged over wall translates, is at most

$$C\|z\|_\infty^2(R/P)\#\{\text{active observation cells}\},$$

by [12, Equation (9.31)]. The other imaginary Gaussian shapes cancel those of the proxies. Real parts are bounded on good boxes by the separate positive-topology estimate (5.21), and on bad boxes by Lemma 5.3. Apply the preceding Gaussian completion to these bounds, including the explicitly charged $CR^{-12}P^2N_{\text{good}}$ term in (5.24); links retain their gains. For fixed strip height H we obtain

$$|F_\pi(\psi + (x + iy)f)| \leq \exp\{o(R)(j_1 + j_2) + C_H(1 + x^2)|X|\} \prod_{\ell} C e^{-c(R+d_\ell)} W_m^\kappa(X, \psi) e^{V_0(X, \psi)}, \quad (5.27)$$

$$|y| \leq H.$$

In particular, the wall charge per halo is $O(R^3/P) = O(R^{-2})$, not $O(R^2)$. Absolute Gaussian integration and analyticity follow from (5.9) with reserve and these strip bounds.

Choose $r_0 > 4eh_0$, then $H > 4r_0$, before R . Apply three-lines on the upper and lower strips to $F_\pi(\psi + zf)e^{-D_0 z^2|X|}$, with $D_0 > C_H$. The real boundary has (5.25) and the real-translation bound; the other boundary has (5.27). On $|z| \leq r_0$ the real-boundary harmonic weight is at least $3/4$, so the exponential item gains survive with a smaller constant. Removing the damping costs $e^{C|X|}$, absorbed because

$$|X| \leq C(j_1 + j_2) + C \sum_{\ell} (1 + d_\ell/R).$$

Cauchy's estimate and real polarization bound the order- r contribution to the analytic point norm by the pattern bound times $(h_0/r_0)^r r^r/r!$. Its sum converges. Thus (5.25), with weaker constants, holds in the full analytic point norm.

Uniformity and completion of entry. At fixed R, P , choose a sufficiently large t , then a sufficiently large s_0 depending on t , using Lemma 5.2 on the finitely many bounded geometries, real-data compact sets, and imaginary-direction compact sets just used. All estimates have strict slack. Fixed-graph sums and conditional bounded integrals are continuous in $\tilde{b}$. Indeed guard dimension is now finite and fixed. Although guard values range over an unbounded lattice, $\delta_i \leq 1$, and the joint laws and fixed tilts have uniform spanning-forest exponential domination on the

compact parameter and observation sets. Their positive normalizing sums have positive compact lower bounds. Dominated convergence therefore applies to the integrated absolute differences; no pointwise uniform estimate at arbitrary guard data is needed. Hence a common real neighborhood of b preserves these bounded-data estimates. The global large-field bound already has a uniform reserve. The same Gaussian averaging and interpolation therefore give the full norm bound throughout this neighborhood; no continuity assertion about the unknown $a(\tilde{b})$ has entered.

Finally use the connected pattern summation of [12, Equation (9.34)]. A halo has bounded block diameter and only polynomially many observation-cell labels per block. A link of length d covers at most $C(1 + d/R)$ blocks. The exponential item and link gains pay all activity weights, coverings, and rooted-tree sums, leaving Ce^{-cR} in the norm. Choose R large enough for (5.13). Absolute estimates justify every exchange with Gaussian integration. Finite-range covariance factorization gives the exact subset algebra, including its exclusions. The covariant tile rules and aligned added form preserve the stated symmetries. The activity hypotheses preceding [12, Proposition 8.3] require block translations and square symmetries, not unit-site translation invariance of the physical interaction; the nested grids have $s_0 \mid s \mid m$ and common square centers. Winding links keep their full covers; every unwrapped padded support therefore has exactly its plane activity. This proves the lemma. $\square$

5.4 Uniform gapped stability and a torus diagnostic

We finish the proof of Proposition 5.1. Apply [12, Proposition 8.3] with the cosine coordinate omitted, which is permitted because $a' > 8\pi$. The reference kernel is fixed and nearest-neighbor; take its starting scale sufficiently large. The modified regulator is allowed by [12, Lemma 9.2]. All map hypotheses now concern the reference and the small activities supplied by Lemma 5.4.

We record why the permitted smallness can be chosen *before* the large entry scale. The Gaussian-curve construction [12, Lemma 10.3] gives exact trajectories by initializing a small multiple d of the block gradient energy and no lattice factor. Their expansion has a linear gradient singleton and an $O(d^2)$ remainder. These estimates are uniform over sufficiently large starting scales: scaled block gradient energy has a uniform point norm, and its small exponential is absorbed by the same regulator reserve. Terminal Gaussian completion is also uniform, because the tail covariance contracts the kinetic energy and the terminal torus has a fixed number of blocks. On a fixed complex disk, $(\text{Id} - dC^{1/2}AC^{1/2})^{-1}$ is uniformly bounded, and determinants cancel in the shifted ratios. The map contraction and terminal gradient diagnostic then prevent a first exit on that disk, exactly as in the proof of [12, Lemma 10.3]. Its analytic inverse disk is consequently uniform in the starting scale as well as the RG step. This uniformity is a consequence of that proof and [12, Lemma 9.2], rather than an extra physical assumption on the strengthened model.

Use the exact Gaussian parameter u for the gradient coordinate and subtract the corresponding Gaussian remainder, writing the transverse remainder as B_{rem} . The balanced linearization is contracting transversely and the pure Gaussian curve has no drift. The analytic map estimates give

$$\|B_{\text{rem},+}\| \leq [\theta + C(|u| + \|B_{\text{rem}}\|)]\|B_{\text{rem}}\|, \quad (5.28)$$

$$|u_+ - u| \leq C(|u| + \|B_{\text{rem}}\|)\|B_{\text{rem}}\|. \quad (5.29)$$

These are the fixed-reference inequalities [12, Equation (10.10)] and [12, Equation (10.11)]. Choose a small tube so that the first factor is at most $\theta' < 1$. Small enough entry data stay in that tube, with $\|B_{\text{rem},j}\| \leq (\theta')^j \|B_{\text{rem},0}\|$ and summable gradient drift. In particular all activities remain as small as desired. The separate torus remainder obeys the same uniform small bound by its contraction with quadratic forcing and the exact copying of singleton coefficients in [12, Proposition 8.3].

Choose a sufficiently large fixed dyadic D for the causal neighborhoods of these maps and use compatible tori of side

$$n_j = DmL^j. \quad (5.30)$$

Take j large enough that $n_j \geq sP$, so the exact entry is available. At the stopping scale there are D^2 blocks. Let f be a nonzero mean-free first cosine on the unit torus, and let f_n be its area-weighted observation-cell samples. Put

$$g = \langle f, (-\Delta_{\mathbb{T}^2})^{-1} f \rangle > 0.$$

The reference variance of $\langle Y, f_n \rangle$ tends to $a'g$. Gaussian completion of a smooth real source translates the exact relative density by the reference covariance of that source. The Gaussian Laplace prefactor remains, while the fixed terminal block count and regulator reserve bound the differentiated logarithmic correction by the small terminal activity norms. This is the gradient diagnostic [12, Equation (10.3)]. It requires only smallness here, not convergence of the gradient coordinate to zero and not a derivative in $\tilde{b}$. By choosing the tube small, for all sufficiently large j ,

$$\text{Var}_{\text{strengthened}} \langle Y, f_{n_j} \rangle \geq \frac{1}{2} a'g. \quad (5.31)$$

The observation noise has variance $\sum_x f_{n_j}(x)^2 = O((s/n_j)^2)$, which tends to zero. Removing the added positive precision increases centered height variance by Lemma 2.2. Hence the original periodic Skellam height law satisfies

$$\liminf_{j \rightarrow \infty} \text{Var}_{b, \text{tor}}^{\sim} \langle h, B_s^* f_{n_j} \rangle > 0 \quad (\tilde{b} \in I). \quad (5.32)$$

If $a(\tilde{b}) = 0$, the left side must instead vanish. To see this directly, first take a smooth periodic gradient test. Split the torus into macroscopic rectangles, release the bonds between them, and retain the internal gradient observations. For a fixed macroscopic partition there are $O(n)$ seam edges and their coefficients are $O(n^{-1})$, so the discarded seams have reference cost $O(n^{-1})$; the free-cell variances converge to zero by Theorem 2.4. The variance comparison and Lemma 2.3 give vanishing variance for the original periodic test. The same conclusion holds for smooth mean-zero density tests by using the periodic inverse Laplacian and reference-energy approximation. Finally, the observation-cell profile $B_s^* f_n$ approximates the corresponding smooth density in reference cost as $n \rightarrow \infty$, with reference-cost error $O((s/n)^2)$, since s is fixed. Equivalently one may use the sampled periodic cosine shift and estimate its cell-averaging error. Thus its variance also tends to zero, contradicting (5.32).

It follows that $a(\tilde{b}) > 0$ throughout the real neighborhood I . Shrinking I inside $(0, \infty)$ proves Proposition 5.1.

6 Identification of the critical exponent

Proof of Theorem 1.1. Apply Proposition 2.6 at $b = b_c$. If $a(b_c) = 0$, it gives

$$\limsup_n \frac{\log C_{b_c}(n)}{\log n} = -\infty,$$

whereas Proposition 1.2 gives a liminf of at least -1 . Hence $a(b_c) > 0$, and Theorem 2.5 implies

$$a(b_c) \geq 8\pi.$$

Suppose the inequality were strict. Proposition 5.1 would give $a(\tilde{b}) > 0$ for all $\tilde{b}$ in a neighborhood of b_c . Since $b_c > 0$, this neighborhood contains some $0 < \tilde{b} < b_c$. Proposition 4.1 then gives a

polynomial lower bound for $C_{\tilde{b}}(n)$, and therefore $m(\tilde{b}) = 0$: indeed $C_{\tilde{b}}(n) \leq 1$, and its negative logarithm is $O(\log n)$. This contradicts (1.3). We have proved

$$a(b_c) = 8\pi.$$

Propositions 2.6 and 4.1 now give

$$-\frac{1}{4} \leq \liminf_n \frac{\log C_{b_c}(n)}{\log n} \leq \limsup_n \frac{\log C_{b_c}(n)}{\log n} \leq -\frac{1}{4}.$$

Both estimates use the free-box limit first and then all integer separations, so they establish exactly the limit in the theorem. The quantified bounds follow from the definition of this limit. $\square$

The conclusion does not assert convergence of $n^{1/4}C_{b_c}(n)$. In particular, multiplicative logarithmic corrections are compatible with the result. The proof claims rest on the arguments and cited results presented here.

References

- [1] M. Aizenman, M. Harel, R. Peled, and J. Shapiro. Depinning in integer-restricted Gaussian fields and BKT phases of two-component spin models. [arXiv:2110.09498v3](#), May 27, 2022; first posted in 2021.
- [2] V. L. Berezinskii. Destruction of long-range order in one-dimensional and two-dimensional systems having a continuous symmetry group. I. Classical systems. *Soviet Physics JETP* **32**(3) (1971), 493–500. Russian original: *Zh. Eksp. Teor. Fiz.* **59** (1970), 907–920. [English translation](#).
- [3] T. Durand and P. Lammers. The BKT transition and surface tension differentiability. [arXiv:2605.28473v1](#), May 27, 2026.
- [4] J. Fröhlich and T. Spencer. The Kosterlitz–Thouless transition in two-dimensional Abelian spin systems and the Coulomb gas. *Communications in Mathematical Physics* **81**(4) (1981), 527–602. [doi:10.1007/BF01208273](#).
- [5] J. Ginibre. General formulation of Griffiths’ inequalities. *Communications in Mathematical Physics* **16**(4) (1970), 310–328. [doi:10.1007/BF01646537](#).
- [6] J. M. Kosterlitz. The critical properties of the two-dimensional XY model. *Journal of Physics C: Solid State Physics* **7**(6) (1974), 1046–1060. [doi:10.1088/0022-3719/7/6/005](#).
- [7] J. M. Kosterlitz and D. J. Thouless. Ordering, metastability and phase transitions in two-dimensional systems. *Journal of Physics C: Solid State Physics* **6**(7) (1973), 1181–1203. [doi:10.1088/0022-3719/6/7/010](#).
- [8] P. Lammers. Bijecting the BKT transition. [arXiv:2301.06905v2](#), April 18, 2023.
- [9] P. Lammers. A dichotomy theory for the height functions of the BKT transition. [arXiv:2211.14365v4](#), July 25, 2026; first posted in 2022.
- [10] T. Lupu. From loop clusters and random interlacements to the free field. *The Annals of Probability* **44**(3) (2016), 2117–2146. [doi:10.1214/15-AOP1019](#).

- [11] O. A. McBryan and T. Spencer. On the decay of correlations in $SO(n)$ -symmetric ferromagnets. *Communications in Mathematical Physics* **53**(3) (1977), 299–302. doi:10.1007/BF01609854.
- [12] OpenAI. BKT universality for height and planar spin fields. OpenAI Math Release preprint OAI:BKT-universality-for-height-and-planar-spin-fields-September-24-2026, 2026. All numbered companion citations refer to this paper.
- [13] O. Regev and N. Stephens-Davidowitz. An inequality for Gaussians on lattices. *SIAM Journal on Discrete Mathematics* **31**(2) (2017), 749–757. doi:10.1137/15M1052226. arXiv:1502.04796v3.
- [14] D. van Engelenburg and M. Lis. An elementary proof of phase transition in the planar XY model. *Communications in Mathematical Physics* **399**(1) (2023), 85–104. doi:10.1007/s00220-022-04550-3. arXiv:2110.09465v2. The multiplicative correlation inequality is cited from Lemma 19 of the appendix to this arXiv version; that appendix is absent from the journal article.
- [15] D. van Engelenburg and M. Lis. On the duality between height functions and continuous spin models. *Probability and Mathematical Physics* **6**(4) (2025), 1291–1325. doi:10.2140/pmp.2025.6.1291.